A Preference for Traceable Agency

An Axiomatization of Expected Utility plus Mutual Information*

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Abstract

An action can matter twice: through the payoff it produces and the visible trace it leaves. A channel maps actions into consequences; preferences rank action lotteries in each channel. Data-processing conditions and a branchwise sure-thing principle, with standard axioms, characterize expected utility plus λ times the mutual information between action and consequence. The theorem identifies λ, the price of trace in payoff units; expected utility is the case $\lambda = 0$. The criterion reads the coupling of act and consequence, not consequences alone; randomisation can be strictly optimal. When each action leaves a distinct trace, the optimal lottery has multinomial-logit probabilities; when traces overlap, substitution depends on the similarity of their consequence distributions and independence of irrelevant alternatives can fail.

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1 Introduction

Shower upon him every earthly blessing ... He would even risk his cakes and would deliberately desire the most fatal rubbish, the most uneconomical absurdity ... simply in order to prove to himself—as though that were so necessary—that men still are men and not the keys of a piano. ... If you say that all this, too, can be calculated and tabulated ... then man would purposely go mad in order to be rid of reason and gain his point!

— Fyodor Dostoevsky, *Notes from the Underground*, Part I, Chapter VIII [21]

Dostoevsky's Underground Man faces a choice problem. The cakes are material payoff, and he will risk them so that his conduct can attest to being his own, not the mechanical implication of a table. Yet the proof he seeks is self-defeating. An act fixed in advance cannot provide it, and deliberate rebellion, too, can be "calculated and tabulated." So he would "purposely go mad in order to be rid of reason"—not because madness is what he values, but because it is his last imagined way of preventing the table from closing before he acts. This paper takes the evidentiary idea literally. In Dostoevsky's example, conduct is observed directly: the act supplies its own evidence. We consider the general case, in which an act is observed only through a possibly noisy payoff-bearing outcome and an explicit visible record.

For visible consequences to carry information about an act, two conditions are needed. There must be uncertainty beforehand about which act will occur, and the consequence must discriminate among the possible acts. Either condition can fail on its own. A lottery over acts leaves no trace when every act generates the same distribution of visible consequences; a consequence that perfectly reveals the act conveys nothing new when the act was already known. We call the intrinsic preference for the two together a preference for *traceable agency*. The paper axiomatizes it jointly with ordinary preferences over payoff risk.

Let A be a finite set of actions, O a finite set of payoff-bearing outcomes, R a finite set of explicit visible records, and $K : A \to \Delta(O \times R)$ the channel relating actions to visible consequences. An action lottery $q \in \Delta(A)$ gives the distribution of the realised action. The primitive is a family $\{\succeq_K\}_K$, with one binary relation $\succeq_K$ on $\Delta(A)$ for each fixed channel K. The statement $q \succeq_K p$ means that q is weakly preferred to p in environment K. Eight axioms are imposed on this family. To compare lotteries governed by different channels, the channels are placed side by side as blocks of one larger environment; duplicating a channel or adding an unused block does not alter the comparison at issue. Two processing axioms say that the agent never gains when her record is garbled or her action variable is processed noisily, and a benchmark comparison rules out indifference to trace. A sure-thing principle for records that arrive in stages supplies the cardinal structure. No audience appears in the model, and the taste is content-neutral: it prices how much the consequence reveals, not what it reveals.

The main result is an if-and-only-if characterization. A family $\{\succeq_K\}_K$ satisfies the eight axioms if and only if there are a nonconstant payoff utility $u : O \to \mathbb{R}$ and $\lambda > 0$ such that, for every finite joint channel K and every $q, p \in \Delta(A)$,

$$q \succeq_K p \quad \Longleftrightarrow \quad \mathbb{E}_{qK}[u(O)] + \lambda I_{qK}(A; O, R) \geq \mathbb{E}_{pK}[u(O)] + \lambda I_{pK}(A; O, R),$$

where $I_{qK}(A; O, R)$ is the mutual information between the action and its visible consequence. We call such families *trace-tempered*. The utility u is unique up to positive affine transformations, λ is measured in payoff units per unit of information, and the same pair prices comparisons across channels, embedded as blocks of one larger environment. A sign trichotomy completes the picture: reversing the axioms that carry the taste for trace characterizes $\lambda < 0$—a preference for deniability—while weakening them to indifference yields expected utility, $\lambda = 0$. Equivalently, the information term prices control: how strongly the distribution of visible consequences depends on the realised act. Trace and control are not separate components of preference but two descriptions of the same act–consequence dependence. No choice among lotteries separates them.

The information term has a simple interpretation. Observing (O, R) induces a posterior belief about the realised action; traceability is the expected reduction in that uncertainty, measured by Shannon entropy H. If visible consequences are independent of actions, the value is zero. If the action is known in advance because q is degenerate, the value is also zero. If each action in the support of q produces a distinct visible consequence, the value is $H(q)$—Dostoevsky's case, in which conduct is observed directly.[1] Thus the criterion is not a taste for randomisation as such. Randomisation matters only because it creates uncertainty about agency for the visible consequence to resolve.

Traceable agency is thus a taste for the coupling of act and consequence, not for consequences alone. A four-action channel turns this into a direct test. Every action pays the same amount and writes one of two symbols on the record. Action a_1 always writes the first symbol and a_2 the second; a_3 and a_4 each write either symbol with equal probability. The choice is between a fair coin over $\{a_1, a_2\}$ and a fair coin over $\{a_3, a_4\}$. The two coins randomise identically and induce the same joint distribution of payoff and record. Under the first coin the record identifies the realised action; under the second it reveals nothing. Any criterion that reads only the distribution of consequences is indifferent between the coins; so is a taste for randomisation as such. Traceable agency predicts a strict preference for the first. Because the record affects neither payoffs nor any later decision, the preference cannot be instrumental.[2] The comparison identifies a value of act–consequence dependence; by itself it does not distinguish trace from control or identify the psychological source of that value.

This comparison has not been run. The nearest evidence comes from two experiments. Bartling, Fehr, and Herz elicit each principal's point of indifference between retaining and delegating the right to choose a project and the effort devoted to it. At that point, the delegation lottery has a certainty equivalent 16.7% higher on average than the lottery produced by the principal's own choices, when both are valued outside the delegation setting [7]. This is consistent with a preference for trace, since retaining control makes the principal's choices determine the payoff lottery, but it may instead reflect autonomy or some other value of deciding for oneself. Mistry and Liljeholm, and later Norton and Liljeholm, ask subjects to select a room containing two slot machines and then gamble in that room. Subjects are more likely to choose a high-divergence room over a zero-divergence room when they choose between its machines themselves than when a computer chooses for them, holding current expected payoffs and outcome entropies fixed [37, 38]. With equal weight on the two machines, the divergence between their token distributions is the mutual information between the machine and the token. This also admits a trace interpretation: in self-play, the token is evidence about a voluntarily selected machine. But token values could change after the room was selected, allowing subjects in self-play to adapt their machine choices. Thus trace is confounded with the value of deciding in the first design and with that of adapting in the second. The test proposed above removes both confounds.

Section 2 sets out the domain and the eight axioms. Section 3 states the representation theorem, with a proof sketch, and derives uniqueness, the sign trichotomy, and the independence of the axioms. Section 4 presents a two-treatment test that can reject the model; derives deliberate randomisation, with multinomial logit as the boundary case when every action leaves its own mark; prices garblings of the record; measures λ from choices; and characterises when one agent is more trace-seeking than another. Section 5 reviews the literature, in particular rational inattention, where mutual information is a cost of attention rather than an object of taste, and the axiomatic characterisations of information indices. Section 6 discusses interpretation and scope. The proofs are in the appendices: the pure-trace characterisation in Appendix A, the completion of Theorem 1 in Appendix B, and the remaining results in Appendix C.

[1] If the visible-consequence supports of all channel rows are pairwise disjoint, the unique optimal lottery has multinomial-logit weights. Overlap can make independence of irrelevant alternatives fail; see Remark 9.

[2] A second treatment completes the design: repeat the choice in a channel that writes a fair record independently of the realised action. Every representation then assigns both flips the same value, whatever u and λ, so a strict preference there rejects the model. Section 4 develops the test and what it excludes.

2 The model

Fix a finite set O of *payoff-bearing outcomes*, with $|O| \geq 2$. Let A be a nonempty finite set of *actions* and R a nonempty finite set of *explicit visible records.* A joint channel is a stochastic kernel

$$K : A \longrightarrow \Delta(O \times R).$$

For each finite joint channel K, the primitive behavioural datum is a binary relation

$$\succeq_K \quad \text{on} \quad \Delta(A).$$

The interpretation is that q is a lottery over actions used by the agent and $q \succeq_K p$ means that, in the fixed environment K, q is evaluated at least as highly as p. Within that environment, the lottery is the procedure being evaluated and the channel is fixed.[3]

Axioms are imposed on the family $\{\succeq_K\}_K$. The first two axioms are standard, and Axiom (A3) is the usual nondegeneracy condition on material outcomes. Axioms (A4), (A6), and (A7) carry the preference for trace. Axiom (A5) puts the separate within-channel rankings on a common scale, while Axiom (A8) imposes consistency across stages; neither fixes the sign of the trace term.

(A1) Weak order. For every finite joint channel K, $\succeq_K$ is complete and transitive on $\Delta(A)$.

(A2) Continuity. For each fixed triple of alphabets (A, O, R), the set

$$\{(K, q, p) : q \succeq_K p\} \subseteq \Delta(O \times R)^A \times \Delta(A)^2$$

is closed.

(A3) Material relevance. There are distinct outcomes $o^+, o^- \in O$ and a two-action channel K with no record in which a^+ delivers o^+ for sure and a^- delivers o^- for sure. For this channel,

$$\delta_{a^+} \succ_K \delta_{a^-}.$$

The agent strictly ranks at least two outcomes when each is delivered for sure and leaves no record.

(A4) Trace relevance. There is an outcome $o^* \in O$ and a four-action channel K in which every action delivers o^* for sure. Actions a_1 and a_2 leave distinct records r_1 and r_2, respectively, while each of a_3 and a_4 leaves either record with equal probability. For this channel,

$$\frac{1}{2}(\delta_{a_1} + \delta_{a_2}) \succ_K \frac{1}{2}(\delta_{a_3} + \delta_{a_4}).$$

Both are fair lotteries and produce the same distribution of visible consequences. The axiom requires a strict preference for the first, whose record identifies the realised action, over the second, whose record does not.

[3] Why not a single preference over lotteries on $A \times O \times R$? Every joint law is $q \otimes K$ for some pair, so the domain is rich enough. But in any one choice situation the channel is fixed: the agent moves the marginal on A, never the coupling. A preference over all joint laws would rank couplings, and no choice reveals those rankings.

Comparison environments. Some axioms compare lotteries carried by different channels. Each relation $\succeq_K$ ranks lotteries within one channel, so such comparisons need a domain in which two channels appear in a single preference statement. A larger channel that contains both as labelled blocks provides it, much as objective lotteries enrich Savage's domain in Anscombe and Aumann [4].

For $K : A \to \Delta(O \times R)$ and $L : B \to \Delta(O \times R_L)$, let $K \sqcup L$ be the block-diagonal joint channel—the *block environment*—with action set $(A \times \{0\}) \cup (B \times \{1\})$, payoff set O, and record set $(R \times \{0\}) \cup (R_L \times \{1\})$: each block operates its own channel, and all cross-block probabilities are zero. For $q \in \Delta(A)$ and $p \in \Delta(B)$, let q^0 and p^1 denote their block-supported copies, defined on the tagged action set by

$$\begin{aligned} q^0(a,0) &:= q(a), & q^0(b,1) &:= 0, \\ p^1(a,0) &:= 0, & p^1(b,1) &:= p(b), \end{aligned} \qquad (a \in A,\ b \in B).$$

Under q^0 the visible consequence is generated by K, and under p^1 by L, so a choice between them is also a choice between the two channels, made inside a single environment. When the blocks have the same action alphabet, q^0 and q^1 are the two tagged copies of the same lottery. Write

$$(q, K) \succeq (p, L) \iff q^0 \succeq_{K \sqcup L} p^1.$$

Each instance of this shorthand is a single comparison inside the two-block environment $K \sqcup L$, which is built from the two channels; $\succ$ and $\sim$ denote the asymmetric and symmetric parts of this shorthand relation. The pair (q, K) has no meaning in isolation.[4] The construction extends to any finite family $\{K_i : A_i \to \Delta(O \times R_i)\}_{i \in I}$: write $\bigsqcup_{i \in I} K_i$ for the analogous block environment and q_i^i for the lottery that assigns $q_i(a)$ to (a, i) and zero to every other block.

(A5) Block-comparison coherence. *Duplication:* for every $K : A \to \Delta(O \times R)$ and all $q, p \in \Delta(A)$,

$$q \succeq_K p \iff (q, K) \succeq (p, K).$$

Irrelevant blocks: for every finite block environment $\bigsqcup_{k \in I} K_k$, every ordered pair of distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff (q_i, K_i) \succeq (q_j, K_j).$$

Comparing q and p across two tagged copies of K is just comparing them in K; adding blocks not involved in a comparison leaves it unchanged.

Processing. The next two axioms use two ways of transforming an environment. A *record processor* is a stochastic kernel $T : O \times R \to \Delta(O \times R')$ that garbles the record without changing the payoff: $T(o', r' \mid o, r) = 0$ whenever $o' \neq o$. The garbled channel is simply the matrix product $KT : A \to \Delta(O \times R')$,

$$(KT)(o', r' \mid a) := \sum_{o,r} K(o, r \mid a) T(o', r' \mid o, r).$$

An *action processor* is a stochastic kernel $S : A \to \Delta(B)$ that replaces the action variable A by a processed action variable B. Given $q \in \Delta(A)$, it induces the lottery $qS \in \Delta(B)$,

$$(qS)(b) := \sum_a q(a) S(b \mid a).$$

[4]Formally, $(K \sqcup L)(o, (r, 0) \mid (a, 0)) = K(o, r \mid a)$ and $(K \sqcup L)(o, (r_L, 1) \mid (b, 1)) = L(o, r_L \mid b)$, with every other entry zero. In the binary case, with $A = \{a_1, a_2\}$ and $B = \{b_1, b_2\}$, the block environment has the four tagged actions $(a_1, 0)$, $(a_2, 0)$, $(b_1, 1)$, $(b_2, 1)$; a lottery $q = (q(a_1), q(a_2))$ becomes $q^0 = (q(a_1), q(a_2); 0, 0)$ and $p = (p(b_1), p(b_2))$ becomes $p^1 = (0, 0; p(b_1), p(b_2))$, the semicolon separating the blocks. Then $(q, K) \succeq (p, L)$ abbreviates $q^0 \succeq_{K \sqcup L} p^1$: one ordinary comparison of two lotteries in this one four-action environment.

A *Bayesian pushforward* of K by S at q is any channel $\widehat{K} : B \to \Delta(O \times R)$ satisfying[5]

$$(qS)(b)\,\widehat{K}(o,r \mid b) = \sum_a q(a)S(b \mid a)K(o,r \mid a) \quad \text{for every } (b,o,r). \tag{1}$$

In the processed environment, the alternatives are the noisy names b, while the visible consequences remain (O,R). Choosing qS under $\widehat{K}$ replaces dependence on the original action with dependence on its noisy name, without changing the marginal distribution of visible consequences.

(A6) Record data processing. For every $K : A \to \Delta(O \times R)$, every record processor T, and every $q \in \Delta(A)$,

$$(q,K) \succeq (q,KT).$$

The agent never gains when her record is garbled.

(A7) Action data processing. For every $K : A \to \Delta(O \times R)$, every action processor $S : A \to \Delta(B)$, every $q \in \Delta(A)$, and every Bayesian pushforward $\widehat{K}$ of K by S at q,

$$(q,K) \succeq (qS,\widehat{K}).$$

Nor does she gain when the action variable is processed, holding the distribution of visible consequences fixed.

Compounding. A first-stage record channel $P : A \to \Delta(Y)$ and a *profile of continuation channels* $K^y : A \to \Delta(O \times R_y)$ compose into the *compound environment*

$$(P \triangleright \{K^y\})(o,(y,r) \mid a) := P(y \mid a)K^y(o,r \mid a). \tag{2}$$

In words, one action is realised and the environment responds in two stages. First, the action a generates only an interim record y through P; no payoff is delivered at this stage. Then the same action generates the payoff o and a further record r through K^y. The agent does not act again. The interim record remains visible, so the final record is (y,r). Given an action lottery q, let q_y denote the posterior distribution of the action after an interim record y that occurs with positive probability.[6]

(A8) Branchwise continuation consistency. For every finite family of record alphabets $(R_y)_{y\in Y}$, every first-stage record channel $P : A \to \Delta(Y)$, every pair of continuation profiles $K^y, L^y : A \to \Delta(O \times R_y)$, and every $q \in \Delta(A)$: if

$$(q_y,K^y) \succeq (q_y,L^y) \quad \text{for every } y \text{ that occurs with positive probability under } q \text{ and } P,$$

then

$$(q,\ P \triangleright \{K^y\}) \succeq (q,\ P \triangleright \{L^y\}),$$

strictly if one such branch comparison is strict.

The axiom holds the action lottery q and the first-stage record channel P fixed. The interim record may change the posterior over actions and hence the conditional ranking of the branches. But if, conditional on every interim record y that can occur, the branch governed by K^y is weakly preferred to the branch governed by L^y, then their common first stage cannot reverse that ranking: the compound using $\{K^y\}$ is weakly preferred to the compound using $\{L^y\}$. The preference is strict if one such conditional preference is strict. Zero-probability branches are unrestricted, exactly as Savage's null events are. The branching event is always a record, never a payoff.

When P is an action-independent public coin, Axiom (A8) reduces to ordinary mixture independence.

[5]If $(qS)(b) = 0$, the equation leaves that row unrestricted; every statement below applies to every consistent completion, making unreached processed actions irrelevant.

[6]Thus $q_y(a) := \frac{q(a)P(y|a)}{\sum_{a'} q(a')P(y|a')}$ whenever $\sum_{a'} q(a')P(y \mid a') > 0$.

3 The representation theorem

For $q \in \Delta(A)$ and $K : A \to \Delta(O \times R)$, let

$$p_{qK}(o, r) := \sum_a q(a) K(o, r \mid a)$$

be the induced distribution of visible consequences, and define

$$I_{qK}(A; O, R) := \sum_{a,o,r} q(a) K(o, r \mid a) \log \frac{K(o, r \mid a)}{p_{qK}(o, r)}. \tag{3}$$

Summands with $q(a)K(o, r \mid a) = 0$ are set to zero, the logarithm has any fixed base greater than one, and expectations below are taken under the joint distribution $q(a)K(o, r \mid a)$.

Theorem 1. *For a family $\{\succeq_K\}_K$ as described above, the following are equivalent.*

(i) *The family satisfies Axioms* (A1)–(A8).

(ii) *There are a nonconstant function $u : O \to \mathbb{R}$ and a number $\lambda > 0$ such that, for every nonempty finite A, R, every $K : A \to \Delta(O \times R)$, and every $q, p \in \Delta(A)$,*

$$q \succeq_K p \iff V(q, K) \geq V(p, K), \qquad V(q, K) := \mathbb{E}_{qK}[u(O)] + \lambda I_{qK}(A; O, R). \tag{4}$$

Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k \in I} K_k$, every two distinct blocks $i, j \in I$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff V(q_i, K_i) \geq V(q_j, K_j).$$

We call a family satisfying (A1)–(A8) *trace-tempered*, and likewise the criterion that represents it: the taste for trace is tempered by ordinary payoff utility, at the identified rate λ. The criterion is a single expectation. Under the joint law of act and visible consequence, the agent collects $u(O)$ plus λ times the log-likelihood ratio of the realised consequence under her act against its average under the lottery. Payoff and trace enter additively, and the additivity is a claim about behaviour: the agent values what she obtains and what the visible consequence reveals about what she did, not the coincidence of the two. If two channels give the agent the same expected payoff and the same distribution of the posterior about her act, she is indifferent, however each pairs the good payoffs with the revealing records (Corollary 39, Appendix C): there is no premium for the good outcome arriving through the revealing record rather than alongside it. No axiom asserts this separability; it is derived. Compound environments branch formally on interim records. The derivation also uses the richness of the domain: a realised payoff can be disclosed in an interim record and delivered later without changing its value.

3.1 Proof sketch

That the representation satisfies the axioms is a direct verification. The converse has four steps; the complete proof is in Appendix B, resting on a lemma proved in Appendix A.

Step 1: reduction to terminal laws. Fix a full-support q. Two channels that induce the same joint law of realised payoff and posterior about the action are record processings of one another, so Axioms (A1), (A5), and (A6) rank them as indifferent (Lemma 34). Axiom (A7), applied to bijective processors in both directions, makes relabellings neutral; applied to general q, it deletes null actions and extends the identification beyond full support. Each pair (q, K) is thereby identified with its *terminal law*, the joint distribution of realised payoff and posterior.

Step 2: cardinalisation. A public coin is a first stage that reveals nothing, so every branch posterior equals q. A coin between two environments therefore realises the mixture of their terminal laws, and Axiom (A8) applied to such stages makes preference respect that mixing: independence holds over terminal laws (Lemma 34). With continuity, Axiom (A2), the mixture-space theorem then gives an affine representation on each fixed-q fibre, the family of pairs sharing one lottery. All cardinal content enters here; every other axiom is an ordinal condition, imposed environment by environment.

Step 3: two scales. Two special families of environments deliver the two scales. On singleton action sets there is no trace, and the affine representation is expected utility, nonconstant by Axiom (A3); action processors transport this payoff scale across lotteries and alphabets, and block coherence makes it one scale for all environments (Lemma 35). At the constant payoff that witnesses Axiom (A4), there is no payoff motive, and what preference still ranks is the trace: Appendix A identifies this order with mutual information, Axiom (A4) fixes its positive sign, and Lemma 36 puts these comparisons on the same scale as the payoffs. Here Axioms (A7) and (A8) do different work. Branchwise continuation gives the chain rule, while action processing identifies revelation of an action group with revelation of the corresponding reduced action. Once Axiom (A5) has aligned scales across priors, these implications yield Faddeev's recursion, which characterises Shannon entropy [23].

Step 4: assembly. By Axiom (A8), differences in continuation value aggregate across reached branches. The level of a compound also includes the trace carried by its first-stage record; comparisons with terminal payoffs identify the reached-probability weights (Lemmas 37 and 38). Now sequentialise an arbitrary channel: announce (o, r) as a first-stage record, then deliver o terminally. The result is a record processing of the original, so Axiom (A6) makes the two indifferent, and stagewise assembly gives $\mathbb{E}_{qK}[u(O)] + \lambda I_{qK}(A; O, R)$.

3.2 Uniqueness

To state uniqueness, fix one representation V supplied by Theorem 1. A real-valued index W on all finite lottery–channel pairs is a *common-scale representation* if, for every channel $K : A \to \Delta(O \times R)$ and every $q, p \in \Delta(A)$,

$$q \succeq_K p \quad \iff \quad W(q, K) \geq W(p, K),$$

and, for every finite block environment $\bigsqcup_{k\in I} K_k$, every two distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k\in I} K_k} q_j^j \quad \iff \quad W(q_i, K_i) \geq W(q_j, K_j).$$

Call a common-scale representation W *branch-additive* if, for every first-stage record channel P, every $q \in \Delta(A)$, and every two continuation profiles $\{K^y\}$ and $\{L^y\}$ admitted in Axiom (A8),

$$W\big(q, P \triangleright \{K^y\}\big) - W\big(q, P \triangleright \{L^y\}\big) = \sum_{y:m(y)>0} m(y)\big[W(q_y, K^y) - W(q_y, L^y)\big]. \tag{5}$$

Thus continuation-value differences aggregate by their reached-branch probabilities; stating the requirement in differences avoids fixing an arbitrary zero for material utility.

Proposition 2 (Uniqueness)**.** *Suppose the equivalent conditions of Theorem 1 hold. For any real-valued index W on all finite lottery–channel pairs:*

(i) *W is a common-scale representation if and only if $W = \varphi \circ V$ for a single strictly increasing $\varphi : [\underline{u}, \infty) \to \mathbb{R}$, where $\underline{u} := \min_{o\in O} u(o)$.*

(ii) *A common-scale representation W is branch-additive if and only if $W = \alpha V + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$.*

The proof is in Appendix C.

The primitive remains ordinal: part (i) permits any common increasing transformation. Branch additivity is an accounting requirement on an index, not another behavioural axiom; part (ii) selects the affine gauge. Consequently, if (u, λ) and $(\widetilde{u}, \widetilde{\lambda})$ are two trace-tempered representations, then

$$\widetilde{u} = \alpha u + \beta, \qquad \widetilde{\lambda} = \alpha\lambda$$

for some $\alpha > 0$ and $\beta \in \mathbb{R}$. Once two non-indifferent payoff outcomes are assigned numerical utilities, λ is therefore uniquely measured in those payoff units per unit of information. Unlike in the pure-trace model of Appendix A, its magnitude is not merely a choice of information unit once payoff utility has been calibrated: it is the identified payoff–trace trade-off.

3.3 Sign of the trace weight

Uniqueness concerns scale; a complementary observation concerns sign. Axioms (A4), (A6), and (A7) orient the trace component. Write (A4$^-$), (A6$^-$), and (A7$^-$) for the clauses obtained by reversing their displayed comparisons, and write (A4^0), (A6^0), and (A7^0) for the corresponding indifference clauses.

Corollary 3 (Sign trichotomy)**.** *Suppose Axioms* (A1)–(A3), (A5)*, and* (A8) *hold. Then:*

(i) *under Axioms* (A4), (A6)*, and* (A7)*, the family has the representation in Theorem 1, with* $\lambda > 0$*;*

(ii) *under* (A4$^-$), (A6$^-$)*, and* (A7$^-$)*, it has the same representation with* $\lambda < 0$*;*

(iii) *under* (A4^0), (A6^0)*, and* (A7^0)*, it is represented by expected utility, equivalently by the same formula with* $\lambda = 0$.

Thus the trace component has exactly the positive, negative, and neutral orientations.

The proof is in Appendix C.

Weakened to indifference, the three orientation clauses yield expected utility. Reversed, they yield the same representation with $\lambda < 0$: an agent who prefers her records garbled and her actions coarsened satisfies the other five axioms just as the trace-seeker does. The clauses cannot point in opposite directions: under the other structural axioms, (A4) is inconsistent with (A6$^-$). Support transport identifies the uninformative test lottery with a total garbling of the revealing one, so (A6$^-$) ranks the former weakly above the latter while (A4) ranks it strictly below.

The negative case is quiet in unconstrained choice over lotteries. If $\lambda < 0$, a payoff-maximising pure action is always optimal; more generally, the optimal lotteries are those supported on payoff-maximising actions for which $I_{qK}(A; O, R) = 0$. Deniability is therefore revealed through constrained lotteries or cross-channel comparisons. If the agent must use a fair lottery over two actions in a fully revealing channel, she will surrender up to $|\lambda| \log 2$ in expected-utility units to replace it with a silent channel.

3.4 Independence of the axioms

Proposition 4 (Independence of the axioms)**.** *For each* $i \in \{1, \dots, 8\}$ *there is a family satisfying the other seven axioms and violating Axiom* (Ai).

The eight families are constructed in Appendix C. Three mark the substantive frontier of the system. A conditional-entropy penalty satisfies every axiom except record data processing and values the removal of extraneous record noise. A Gini posterior-reduction criterion satisfies every axiom except action data processing. The satiation criterion $\mathbb{E}[u(O)] + \min\{I(A; O, R), c\}$, for $c > 0$, satisfies every axiom except branchwise continuation consistency and ceases to value trace once c units of information are secured.

4 Choice implications

This section works out what an agent with the representation of Theorem 1 does: which finite designs falsify the model directly (§4.1), when she deliberately randomises (§4.2), what she will pay for changes to the record technology (§4.3), how that price can be measured and compared across agents (§4.4), and how her rankings compare record technologies—and how they do not (§4.5). Throughout, λ is measured in payoff units per unit of information in the fixed base. Proofs for this section, together with full statements of the supporting results that the text presents in summary form, are collected in Appendix C.3; only one-line verifications are kept here.

4.1 The trace test

The functional V reads the joint distribution of the action and its visible consequence. Existing theories of choice read less: the distribution of consequences, the lottery, or both. The following experiment separates them.

In the first treatment the agent chooses between two fair coins over actions. One flips between two actions that leave distinct marks; the other flips between two actions that produce noise. Both coins yield the same payoff for certain and the same fifty–fifty distribution over records; only the first coin's record identifies the realised action. In the second treatment the record technology is disabled: the same choice is offered in a channel where every action produces noise.

Proposition 5 (The trace test)**.** *Four actions pay the same outcome o_0. Two of them leave distinct marks: a_1 always produces record r_1, and a_2 always produces the distinct record r_2. The other two are indistinguishable: a_3 and a_4 each produce r_1 or r_2 with equal probability. Call this channel K, and let q' be the fair coin over $\{a_1, a_2\}$ and q'' the fair coin over $\{a_3, a_4\}$.*

(i) *The two coins induce the same joint distribution of visible consequences, $p_{q'K} = p_{q''K}$; yet*

$$V(q', K) - V(q'', K) = \lambda \log 2 > 0, \qquad \text{so} \qquad q' \succ_K q''.$$

(ii) *Let K° be the channel in which every action pays o_0 and produces r_1 or r_2 with equal probability. Then $q' \sim_{K^\circ} q''$.*

The prediction is strict preference for the revealing coin in the first treatment and exact indifference in the second. Part (i) alone rules out every consequentialist theory. The two coins induce the same distribution over payoff–record pairs, so any criterion that evaluates only that distribution is indifferent between them: expected utility over payoffs, expected utility that values records directly as consequences through some $v(o, r)$, and every non-linear functional of $p_{q'K}$. What separates the coins is the joint distribution of the action with its visible consequence, and that is precisely what consequentialist theories do not read.

Part (i) restricts only how a theory reads consequences. A theory that also reads the lottery can match part (i). Existing theories of deliberate randomisation are of this kind: in the hedging model of Cerreia-Vioglio, Dillenberger, Ortoleva, and Riella [16], and in the preference for randomisation documented by Agranov and Ortoleva [2], the value of mixing depends only on the induced distribution and the lottery. Call an index W *channel-blind* if it reads only these two objects: there are functions F and $h : \Delta(A) \to \mathbb{R}$, fixed across channels, such that

$$W(q, K) = F\big(p_{qK}, h(q)\big).$$

The second treatment rules these out. Nothing a channel-blind index reads changes between the treatments: in all four lottery–channel combinations the distribution of visible consequences is uniform on $\{(o_0, r_1), (o_0, r_2)\}$, and the lotteries are the same two lotteries. A channel-blind index therefore separates the coins in both treatments or in neither, so none reproduces the pattern of Proposition 5: strict preference under K and indifference under K°.[7]

The test also rejects. The first treatment separates the three cases of Corollary 3: the revealing coin if $\lambda > 0$, the other coin if $\lambda < 0$, indifference if $\lambda = 0$. In the second treatment every representation assigns both coins the same value, whatever u and λ, so a strict preference there rejects the model. The lottery should be knowingly chosen and implemented by the subject. Repeating the same statistical lottery under external assignment provides a control for whether the provenance of randomisation matters.

4.2 Deliberate randomisation

A degenerate lottery leaves no trace, so an agent with $\lambda > 0$ may deliberately randomise: she gives up expected payoff to make her act genuinely uncertain, because only then can its consequence reveal it. In a payoff-based theory a sufficiently large payoff advantage makes a single action optimal. Here it need not: an action that leaves a mark no other action can leave is never wholly abandoned, however poorly it pays, because the first grain of mass on it is infinitely informative per unit, and no finite payoff difference outweighs that. The next proposition states this, and gives the exact terms on which a pure action survives: its payoff advantage must cover, for every rival, the price of the rival's distinctiveness from it.

Some notation. Let $Z := O \times R$ and write $K_Z(z \mid a) := K(o, r \mid a)$ for $z = (o, r)$; let

$$\bar{u}(a) := \sum_{o,r} K(o, r \mid a) u(o), \qquad m_{qK}(z) := \sum_{a} q(a) K_Z(z \mid a)$$

be the expected payoff of an action and the distribution of visible consequences induced by a lottery; m_{qK} is the distribution p_{qK} of Section 3, re-indexed by z. The Kullback–Leibler divergence $D_{\mathrm{KL}}(P \,\|\, Q) := \sum_z P(z) \log(P(z)/Q(z))$ between distributions on Z measures how sharply consequences drawn from P stand out against a background of Q; it equals $+\infty$ when P puts mass where Q does not.

Proposition 6 (Exercised distinctiveness). *Let $Z_K := \{z \in Z : K_Z(z \mid a) > 0$ for some $a \in A\}$.*

(i) *Every optimal lottery q^* satisfies $m_{q^*K}(z) > 0$ for every $z \in Z_K$. Consequently, if every action a has a private consequence—some z_a with $K_Z(z_a \mid a) > 0$ and $K_Z(z_a \mid b) = 0$ for all $b \neq a$—then every optimal lottery has full support.*

(ii) *The pure lottery δ_a is optimal if and only if*

$$\bar{u}(b) + \lambda D_{\mathrm{KL}}(K_Z(\cdot \mid b) \,\|\, K_Z(\cdot \mid a)) \leq \bar{u}(a) \qquad \text{for every } b \in A,$$

and uniquely optimal if these inequalities are strict for $b \neq a$. In particular, if some rival reaches a consequence outside the support of $K_Z(\cdot \mid a)$, then δ_a is not optimal, however large its payoff advantage.

With two actions, the threshold below which the agent mixes is a case of Proposition 6(ii). Let $A = \{a, b\}$, write $K_a := K_Z(\cdot \mid a)$ and $K_b := K_Z(\cdot \mid b)$, and let $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$ be the payoff advantage of a. Then δ_a is optimal if and only if $\Delta \geq \lambda\, D_{\mathrm{KL}}(K_b \,\|\, K_a)$. The threshold is directional: what protects the pure action is the detectability of the rival against the background of the pure action's signature. It is infinite when K_b puts mass where K_a does not—the two-action face of Proposition 6(i).

Below the threshold the agent mixes (Proposition 40, Appendix C.3). The optimal mix is unique, and it puts more mass on the inferior action the larger is λ and less the larger is Δ.

[7] If h does not depend on how the actions are labelled—an entropy bonus, say—the first treatment is enough: the permutation that swaps a_1 with a_3 and a_2 with a_4 turns q' into q'' and leaves p_{qK} unchanged, so $W(q', K) = W(q'', K)$. The second treatment is needed for arbitrary h.

Mixing therefore stops exactly at $\Delta = \lambda\, D_{\mathrm{KL}}(K_b \,\|\, K_a)$. Once Δ has been calibrated on the normalized expected-utility scale, the known divergence makes the switching threshold identify λ. At equal payoffs the mix is the capacity-achieving input of the channel, whatever λ: an agent who values trace at all, however little, randomises at zero stakes so as to maximise the information the visible consequence carries about her act. The comparative statics restate the wedge of the trace test (§4.1): randomisation here is instrumental to trace. An uninformative channel supplies no incentive to mix, and, in the binary case, garbling weakly lowers the payoff threshold below which the superior action ceases to be optimal. A taste for mixing as such is channel-blind.

With more than two actions, the same logic is captured by a first-order condition. The objective $q \mapsto V(q, K)$ is concave in q, so an optimum exists and Kuhn–Tucker conditions characterise it: at an optimum, every action in use offers the same total of expected payoff and priced distinctiveness.

Proposition 7 (Distinctiveness condition)**.** *Let $\lambda > 0$. A lottery q^* is optimal if and only if there is a constant c such that*

$$\bar{u}(a) + \lambda D_{\mathrm{KL}}(K_Z(\cdot \mid a) \,\|\, m_{q^*K}) = c \quad \textit{for } a \in \operatorname{supp}(q^*), \qquad \leq c \quad \textit{for } a \notin \operatorname{supp}(q^*).$$

The proposition gives a *distinctiveness premium.* Each action is credited, on top of its expected payoff, with λ times how far its visible consequences depart from the average visible behaviour that the lottery itself generates; at an optimum this total is equalised across the actions in use. An action can therefore receive more probability because it pays more or because its consequences are more distinctive. The condition mirrors rational-inattention first-order conditions with the channel direction and sign reversed [44, 36].

Which lotteries are optimal depends on the coupling of actions to consequences, not on consequences alone; and optimal lotteries need not be unique. Yet every optimal lottery induces the same distribution of visible consequences. The model therefore predicts the record an observer sees, not the mix that generated it.

Proposition 8 (Optimal marginal and multiplicity)**.** *Write $K_a := K_Z(\cdot \mid a)$ and $C^*(K) := \max_{q \in \Delta(A)} V(q, K)$. There is a unique $m^* \in \Delta(Z)$ such that $m_{q^*K} = m^*$ for every optimal lottery q^*. With*

$$\mathcal{A}^* := \{a \in A : \bar{u}(a) + \lambda D_{\mathrm{KL}}(K_a \,\|\, m^*) = C^*(K)\},$$

the set of optimal lotteries is exactly

$$\{q \in \Delta(A) : m_{qK} = m^*,\ \operatorname{supp}(q) \subseteq \mathcal{A}^*\},$$

and the optimum is unique whenever the rows $\{K_a : a \in \mathcal{A}^*\}$ are affinely independent. If actions are partitioned into classes with identical rows, $V(q, K)$ depends on q within each class only through the class's total mass: every redistribution of an optimal lottery within such a class is optimal, and the optimal lotteries are exactly the lifts of the optimal lotteries of the collapsed channel, whose actions are the classes, each carrying the common row; a lift distributes each class's mass arbitrarily within the class.

The proposition's final claim—that moving mass between actions with identical rows changes nothing—is a test. Split an action into two copies with the same row. The record cannot tell the copies apart, so the value, the induced marginal m^*, and the pair's total probability do not depend on how the mass is divided between them. Any model that gives each action its own weight, as multinomial logit does, predicts instead that duplication raises the pair's combined share. Trace-tempered choice predicts exact invariance.

Remark 9 (Logit boundary)**.** Suppose every action leaves its own mark: the supports of the rows are pairwise disjoint, so the visible consequence identifies the action and $I_{qK}(A;O,R) = H(q)$, the Shannon entropy of the lottery. The unique optimum is then multinomial logit,

$$q^*(a) = \frac{\exp(\bar{u}(a)/\lambda)}{\sum_{a'} \exp(\bar{u}(a')/\lambda)},$$

with $1/\lambda$ as the payoff sensitivity (information in natural logarithms; a change to another base greater than one rescales λ). If instead no action leaves any mark—identical rows—every lottery is optimal. Between these extremes there is no closed form, and the ratio of two actions' probabilities depends on the rest of the menu: by Proposition 7, each action is credited for standing out against the average consequence of the mix, and adding an action changes that average. Thus pairwise-disjoint row supports are sufficient for independence of irrelevant alternatives, while overlap permits, but does not imply, its failure. A three-action example shows the mechanism. Let the rows be

$$K_1 = (1,0,0), \qquad K_2 = (0,1,0), \qquad K_3 = (1/2,0,1/2),$$

and let expected payoffs be equal. With only the first two actions, the optimal lottery is $(1/2,1/2)$. Adding the third gives

$$q^* = (1/3,4/9,2/9), \qquad m^* = (4/9,4/9,1/9).$$

Each active row then has divergence $\log(9/4)$ from m^*, while the ratio of the first two choice probabilities falls from 1 to 3/4—the red-bus/blue-bus pattern. Failures of independence in choice data are commonly attributed to correlated tastes or to costly attention. A taste for trace produces them as well. In rational inattention the same logit formula arises for the opposite reason, as the optimal tilting of a chosen channel at a fixed prior over states [36]; here the channel is fixed and the source is chosen.

4.3 The price of being on the record

Theorem 1 represents comparisons across environments on one scale, so the model prices changes to the record technology itself: what is it worth to the agent that her record is coarsened, that a further signal about her is disclosed, that her mixing device is observed? Each answer is an exact amount of information times λ.

The first result answers the first question. Axiom (A6) asserts only a weak preference against garbling; the representation upgrades the axiom to an exact price and characterises when the price is zero.

Proposition 10 (Price of a record garbling)**.** *Let $K : A \to \Delta(O \times R)$, let T be a record processor, and write $L := KT$. Fix $q \in \Delta(A)$ and let (A,O,R,R') carry the joint law $q(a)K(o,r \mid a)T(r' \mid o,r)$. Then*

$$V(q,K) - V(q,L) = \lambda\, I(A;R \mid O,R') \geq 0,$$

and the following are equivalent: (a) *$V(q,L) = V(q,K)$;* (b) *A and (O,R) are conditionally independent given (O,R');* (c) *the two channels induce the same distribution of the posterior about the act, $\mu_{qL} = \mu_{qK}$ in the notation of Corollary 39.*

Rewriting the record costs λ times the information about the action that the original record carried beyond the rewritten one. A garbling is free exactly when it pools only records that already said the same thing about the action: the agent does not pay for gratuitous detail, only for detail that discriminates among her own actions.

The same accounting runs in reverse. Appending any further signal S drawn given (a,o,r) to the record raises $V(q,\cdot)$ by exactly $\lambda\, I(A;S \mid O,R) \geq 0$: more evidence about oneself never hurts. Under the deniability dual of Corollary 3 the sign reverses, and every additional signal about oneself is a cost. Since the visible pair includes the payoff, the total value of trace in any environment is at most $\lambda \min\{H(q), \log(|O|\,|R|)\}$.

One garbling deserves its own statement. Suppose the environment itself is drawn by a coin that is independent of the action. The agent has no taste for or against the draw itself; what she cares about is whether the coin's outcome goes on the record.

Corollary 11 (The recorded coin). *Let $K, L : A \to \Delta(O \times R)$ and $t \in (0,1)$. Let $tK \oplus (1-t)L$ be the public mixture that draws an action-independent coin Y with $\Pr(Y=0) = t$, applies K if $Y = 0$ and L otherwise, and includes the coin in the record; let $M_t := tK + (1-t)L$ be the rowwise mixture, which hides it. Then, for every $q \in \Delta(A)$,*

(i) $V(q,\, tK \oplus (1-t)L) = t\, V(q,K) + (1-t)\, V(q,L)$;

(ii) $V(q,\, tK \oplus (1-t)L) - V(q, M_t) = \lambda\, I(A; Y \mid O, R) \geq 0$, *with equality if and only if A and Y are conditionally independent given (O,R).*

The proof is in Appendix C.3; part (ii) is Proposition 10 applied to the processor that deletes the coin from the record.

Part (i) says the agent has no taste for or against randomising over environments: with the coin on the record, a mixture of environments is worth the mixture of their values (the preference-level counterpart is Lemma 34). Part (ii) prices the coin's visibility. Hiding it costs $\lambda\, I(A; Y \mid O, R)$. The coin says nothing about the action by itself; the cost is positive whenever knowing which environment operated would sharpen what the visible consequences reveal about the action. So the agent prefers her mixing device observed. Theories that value randomisation as such are silent on this margin.

4.4 Measuring the trace weight

The weight λ is a number: payoff units per unit of information (Proposition 2). This subsection shows how choices reveal it. The idea is compensation. An agent will accept a channel that reveals less about her if she is paid enough, and the payment that leaves her exactly indifferent is the price of the lost information. The domain has no money, but it has two outcomes that Axiom (A3) strictly ranks, and a lottery between them can serve as payment: attach to each environment a side branch, reached by a public coin, that pays the better outcome with some probability and the worse one otherwise—probability β for the first environment, β' for the second. Holding β' fixed, the experimenter moves β until the agent is indifferent between the two environments. Because the public side branch carries no information about the action, changing β or β' changes only its payoff. Each compound's information term is its probability of remaining in the original environment times that environment's trace, and the gap $\beta - \beta'$ at indifference converts into payoff units at odds fixed by the design. When the two environments carry the same expected payoff, the converted gap is λ times their information gap—a gap the experimenter fixed in designing the channels, so no information quantity need be estimated. A corner response—the agent cannot be made indifferent with $\beta \in [0,1]$—brackets the price instead of measuring it. The construction and its exact accounting are Proposition 44 in Appendix C.3; a complete two-treatment protocol, with the garbling control built in, is Remark 47 there.

Two priors turn the design into a test of the functional form itself. For a fully revealing channel, the premium for revelation over silence is $\lambda H(q)$. Eliciting that premium at $q = (1/2, 1/2)$ and at $q = (1/4, 3/4)$ therefore predicts the ratio

$$\frac{\log 2}{\frac{1}{4}\log 4 + \frac{3}{4}\log \frac{4}{3}} \approx 1.23.$$

A quadratic posterior-reduction criterion predicts $4/3$. The second prior thus overidentifies λ and separates the Shannon form from nearby dependence indices.

Suppose the agent is indifferent between two environments, one carrying more information about her act and the other paying more. The indifference is a trade: so much payoff given up per unit of information gained. Nothing in the primitive suggests the terms of this trade should travel—they could depend on the prior, on the payoff level, on how much information is already at stake. The next result says they do not: every such trade, anywhere in the domain, occurs at the single rate λ. Write $E_{qK} := \mathbb{E}_{qK}[u(O)]$.

Proposition 12 (Single information price)**.** *If $(q, K) \sim (p, L)$ and $I_{qK} \neq I_{pL}$, then*

$$\frac{E_{pL} - E_{qK}}{I_{qK} - I_{pL}} = \lambda,$$

and λ is the unique number with this property: for every $\lambda' \neq \lambda$ there are finite pairs $(q, K) \sim (p, L)$ with $E_{qK} + \lambda' I_{qK} \neq E_{pL} + \lambda' I_{pL}$.

In the information–payoff plane, indifference is a family of parallel straight lines of slope $-\lambda$. This constancy is the testable fingerprint of Axiom (A8). An agent whose taste for trace satiates—the independence example for that axiom in Proposition 4—has kinked indifference curves, whose measured slope falls to zero beyond the cap. Finding the same slope across designs with different priors or alphabets therefore over-identifies the model; so does agreement between this elicitation and the switching experiment of §4.2, which measures the same λ from optimal behaviour in a single channel.

Indifference is a knife edge: a subject may never report it, and a corner response only brackets the price. Plain choices still bound the weight. Compare two agents facing the same channel, with weights $\lambda_1 < \lambda_2$. The agent who cares more about trace chooses at least as much information and at most as much payoff, and the rate at which the two chosen bundles trade payoff for information lies between the two weights.

Remark 13 (Trace demand)**.** Fix K, write $e(q) := \mathbb{E}_{qK}[u(O)]$ and $i(q) := I_{qK}(A; O, R)$, and let q_j maximise $e + \lambda_j i$ for $0 < \lambda_1 < \lambda_2$. Comparing each optimum at the other weight gives $i_1 \leq i_2$, $e_1 \geq e_2$, and, whenever $i_2 > i_1$,

$$\lambda_1 \leq \frac{e_1 - e_2}{i_2 - i_1} \leq \lambda_2 :$$

the payoff surrendered per additional unit of information brackets the trace weight, whatever selection among optima the data reflect. The indirect value is convex in λ, with the chosen information as its derivative wherever it is differentiable, and its extreme regimes are expected utility ($\lambda \downarrow 0$, with information breaking material ties) and maximal distinctiveness ($\lambda \uparrow \infty$, with payoff breaking information ties). Exact statements are Proposition 41 and Corollary 42 in Appendix C.3.

The comparison of two agents just made presupposed their weights: it ranked λ_1 against λ_2 inside the representation. Choices also settle when that ranking is legitimate—what it means, in behaviour, that one agent is more trace-seeking than another. Call a channel *silent* if every action produces the same distribution of visible consequences; a silent environment offers payoff and nothing else. Say $\{\succeq^2_K\}$ is *more trace-seeking than* $\{\succeq^1_K\}$ if $(q, K) \succeq^1 (p, B)$ implies $(q, K) \succeq^2 (p, B)$ for all lotteries q and p and all channels K and silent B: whenever agent 1 weakly prefers an environment to a silent one, so does agent 2.

Proposition 14 (More trace-seeking)**.** *Let $\{\succeq^1_K\}$ and $\{\succeq^2_K\}$ satisfy Axioms* (A1)–(A8)*. Then $\{\succeq^2_K\}$ is more trace-seeking than $\{\succeq^1_K\}$ if and only if the representations can be chosen with a common utility and $\lambda_2 \geq \lambda_1$.*

The proof is in Appendix C.3. One condition delivers both conclusions. Agreement is forced on the silent environments first—two agents comparable in this sense must share their taste for payoff—and the ordering of the trace weights follows. The comparative is a partial order: agents with genuinely different payoff tastes are incomparable, as in the comparatives of preference for commitment and of ambiguity aversion [29, 25].

4.5 Comparing record technologies

A last question inverts the direction of measurement: not what choices reveal about the agent, but what they reveal about the record technology. Suppose an agent ranks environment K above environment L at *every* lottery. Has she thereby revealed that L is a coarsening of K—that some garbling maps the one record technology into the other? For a Blackwell decision-maker the answer would be yes: dominance in every decision problem is equivalent to garbling [12]. Here the answer is no, and the gap can be stated exactly.

Definition 15. Let $K : A \to \Delta(O \times R)$ and $L : A \to \Delta(O \times R')$ have identical payoff kernels, $\sum_r K(o, r \mid a) = \sum_{r'} L(o, r' \mid a)$ for every (a, o). Write $K \succeq_g L$ if $L = KT$ for some record processor T, and $K \succeq_c L$ if $I_{qK}(A; O, R) \geq I_{qL}(A; O, R')$ for every $q \in \Delta(A)$; the latter is the *more-capable* order on the visible-consequence channels [33].

Proposition 16 (Uniform dominance is more-capable, not garbling)**.** *Let K and L have identical payoff kernels.*

(i) *$K \succeq_g L \Rightarrow K \succeq_c L$, and $K \succeq_c L$ holds if and only if $V(q, K) \geq V(q, L)$ for every $q \in \Delta(A)$.*

(ii) *The converse of the first implication fails: there are channels with identical payoff kernels such that $V(q, K) \geq V(q, L)$ for every q and every representation (u, λ), yet $L \neq KT$ for every record processor T.*

The reason Blackwell's equivalence fails here is that his test family is rich—all state-dependent utilities—while the axioms pin the attitude to records down to the single functional $\mathbb{E}[u] + \lambda I$, so environments are assessed only through mutual information at priors. The revealed informativeness order is then exactly the information theorists' more-capable order, strictly coarser than garbling. And the counterexample is utility-robust—the payoff terms cancel identically—so observing that an agent always ranks one environment above another licenses no inference that the second is a censored version of the first.

The comparison also says where the value of information comes from. For Blackwell's decision-maker information is an input: she can use a finer signal or ignore it, so more is never worse, and monotonicity is a theorem. Here information is an output: the record is about her, and she has no way to discard the evidence her actions generate, because the channel is not hers to process. Whether a finer record is good is therefore not settled by rationality. The axioms settle it by taste: positive for the trace-seeker, negative under the deniability dual of Corollary 3, zero for expected utility. Being better informed is a theorem; being better known is a preference.

5 Related literature

This paper intersects several literatures that differ in what is ranked, which way information flows, and whether an information functional is assumed or derived. In most models of information, a signal tells the agent about an exogenous state; here a visible consequence carries information about the agent's realised action.

Kreps [34] and Dekel, Lipman, and Rustichini [19] use preferences over menus to represent flexibility and uncertainty about future tastes; Pattanaik and Xu [39] rank opportunity sets by the freedom they provide. Gul and Pesendorfer [29] use the same domain to separate temptation, commitment, and self-control, while Kreps and Porteus [35] use temporal lotteries to study preferences over the timing of uncertainty resolution. In this paper, a choice problem fixes the feasible actions, the decision rights, and the channel from actions to visible consequences. The primitive preference ranks lotteries over actions and therefore reads the dependence between the realised action and its visible consequence, rather than the size of a menu, the value of commitment, or the timing of resolution.

Grant, Kajii, and Polak [28] characterise an intrinsic preference for information independent of its use in planning; Caplin and Leahy [15] model anticipatory feelings generated by beliefs, while Ely, Frankel, and Kamenica [22] study preferences over paths of beliefs and the resulting value of suspense and surprise. These papers show that information need not be instrumental, but the relevant beliefs concern uncertainty faced by the agent, whereas the posterior here concerns her realised action. The axioms also restrict its value to expected entropy reduction and fix its rate of substitution with material utility.

Cerreia-Vioglio, Dillenberger, Ortoleva, and Riella [16] study deliberate randomisation and characterise a model in which mixing hedges uncertainty about how to evaluate lotteries; Agranov and Ortoleva [2] document stochastic choice under announced repetition and evidence that much of it is deliberate. Trace-tempered choice can also make a nondegenerate lottery optimal. Proposition 5, however, holds fixed the action lotteries and their distributions of visible consequences while changing whether the consequence identifies the realised action; its second treatment removes that dependence. Together the two treatments rule out every channel-blind criterion that reads only the lottery and its consequence distribution. Randomisation is an implication of traceable agency, not its primitive motive.

In signaling models following Spence [45], actions convey information about hidden characteristics. Glazer and Konrad [27] study charitable giving as a signal of wealth, and Andreoni and Bernheim [3] study social image and audience effects. Bénabou and Tirole [10] allow concerns for social reputation or self-respect. Their earlier paper [9] studies self-serving beliefs about ability, while Bernheim [11] shows how status concerns can induce agents with heterogeneous preferences to choose the same action, making behaviour less informative about predispositions—an instrumental counterpart to the deniability case in Corollary 3. In these models, inference matters through esteem, sanction, reputation, or self-regard. Here no such response enters utility. Beliefs serve only as the statistical yardstick for how clearly the consequence identifies the act.

The proof also relates to axiomatic characterisations of entropy. Shannon [43] characterised entropy from continuity, monotonicity on uniform distributions, and a grouping recursion; Faddeev [23] reduced these to symmetry, continuity of the binary entropy, and the recursion itself. The proof uses the strong-additivity form of Faddeev's theorem given by Baez, Fritz, and Leinster [8]; other formulations and extensions include Khinchin [31], Rényi [41], Aczél and Daróczy [1], and Csiszár [18]. Mutual information can also be obtained through relative entropy, which Baez and Fritz [6] characterise through properties of Bayesian inference.

In those characterisations, a numerical information functional is primitive and structural conditions are imposed directly on it; here the primitive is an ordinal family of weak orders over action lotteries. The information chain rule and Faddeev's recursion are consequences of behavioural axioms, not postulates on a given function. The theorem also calibrates the information term against material utility. The result is not another characterisation of entropy; it derives entropy reduction as the representation of a preference over trace.

Blackwell [12] compares experiments by their value in all downstream decision problems; Gilboa and Lehrer [26] characterise which functions on partitions can arise as values of information in Bayesian decision problems, and Azrieli and Lehrer [5] extend the analysis to stochastic information structures. Cabrales, Gossner, and Serrano [13] obtain entropy reduction from the decisions of ruin-averse investors. In each case, the signal is valuable because it can affect a later decision; here no decision follows the record. Proposition 16 shows that uniform preference between two record technologies identifies the more-capable order, not Blackwell garbling. Being better informed is a theorem; being better known requires an orientation of taste.

The closest formal comparison is with rational inattention and information costs. Sims [44] imposes a finite Shannon-capacity constraint on information flow, while Matějka and McKay [36] show that a cost proportional to mutual information yields a generalized multinomial-logit rule. Rational inattention fixes a prior over states and allows the agent to choose how much to learn before acting, whereas here the channel from actions to consequences is fixed and the agent chooses a distribution over actions. Pairwise-disjoint row supports yield multinomial logit. With overlapping rows, logit can still obtain, but independence of irrelevant alternatives can fail, as Remark 9 shows.

Caplin, Dean, and Leahy [14] characterise Shannon rational inattention through Invariance under Compression and develop posterior-separable generalisations; Denti [20] gives testable conditions under which state-dependent stochastic choice is rationalised by posterior-separable information costs. Frankel and Kamenica [24] characterise measures of information that arise as the value of news in a decision problem and show that the corresponding expected reduction in uncertainty equals expected information. In these papers, the information cost or the underlying optimisation problem supplies the cardinal scale; the present primitive is ordinal, and the common scale for payoff and trace is derived.

Pomatto, Strack, and Tamuz [40] take a cardinal, prior-independent cost functional on Blackwell experiments and show that monotonicity, continuity, product additivity, and constant marginal cost characterise weighted sums of Kullback–Leibler divergences between state-conditional signal distributions. Their costs and the trace value studied here are not nested. The trace component $(q, K) \mapsto I_{qK}(A; O, R)$ depends on the prior and is not a prior-free additive cost of an experiment. Full revelation has finite trace value $H(q)$; under every nonzero cost in their class it is excluded from the finite-cost domain and would have infinite cost. Record and action data processing are behavioural assumptions about whether the agent values evidence of her action; reversing them, with the taste for revelation, yields a preference for deniability, not an information-acquisition cost. The information-cost literature prices learning about the world; the present paper prices being learned about.

6 Interpretation and scope

What the axioms preclude. Mutual information measures how much a record reveals about the act; it assigns no value to which act is revealed. An agent who wants her generous acts recognized and her mean acts hidden may therefore prefer an informative record in one channel and a garbling in another. Such preferences violate the global data-processing requirement in Axiom (A6) and fall outside all three cases of the sign trichotomy. They are not irrational; they are simply not represented here. A model of selective legibility would have to make the value of evidence depend on its content, not only on the amount of uncertainty it removes.

Relation to empowerment. Suppose $\mathbb{E}[u(O) \mid a] = \bar{u}$ for every action. Maximising V over q then gives

$$\max_q V(q, K) = \bar{u} + \lambda C(K),$$

where $C(K) = \max_q I_{qK}(A; O, R)$ is the Shannon capacity of the channel. This is the one-step empowerment index when the visible consequence is interpreted as the agent's sensor state [32, 42]. Empowerment evaluates a channel at its capacity. The representation instead ranks every action lottery q, including those that do not achieve capacity and those for which material utilities differ across actions. At a given q, an action assigned zero probability contributes nothing, even though its availability may raise $C(K)$. Valuing such unused options would require preferences over menus and lies outside the present domain [34].

A belief-based utility, derived. Fix the prior. The trace term becomes a utility over beliefs: visible consequences are valued by the expected reduction in the entropy of the posterior about the agent's act. This is utility from beliefs in the manner of Caplin and Leahy [15]: the posterior enters the objective directly, not as an input to a later decision. Two things distinguish it. The belief here concerns the agent's own conduct, whereas their applications concern external states, such as health. And there beliefs are primitive carriers of utility, free in form; here the belief-based form is derived, and the derivation pins its shape: mutual information priced at one constant rate. Even a taste that satiates after c units of information violates branchwise continuation consistency.

One-shot, with no audience in the model. The model is a single episode: no observer holds the posterior beliefs, no response follows from them, and the staged records in Axiom (A8) belong to one interaction, not to a reputational relationship. The theorem accordingly prices the act–consequence relation itself; it does not identify the psychological source of that value. An experiment could vary whether an audience observes the record, holding the channel fixed: that tests the external-response mechanism—praise, sanction, reputation—but cannot exclude an internal one, such as self-image.

Role of the payoff–record distinction. The payoff–record distinction is needed to identify the representation. First, Axiom (A6) permits the record to be garbled while holding the payoff fixed, thereby restricting the information component without changing payoff utility. Second, Axiom (A8) permits records, but not payoffs, to resolve in stages; this provides the cardinal discipline needed to separate the two components on a common scale. The information term itself treats O and R jointly: either may reveal the act, while only O enters material utility. Once u is normalized, willingness to sacrifice payoff utility for additional mutual information identifies λ.

Proofs

A Pure trace on the constant-payoff domain

The relevance axioms in the main text are stated inside fixed channels. The proofs below use their equivalent comparison-environment forms.

For $o \in O$, write $K_o : \{*\} \to \Delta(O \times \{*\})$ for the channel that delivers o for sure. On a finite action set A, write $K_o^{\mathrm{id}} : A \to \Delta(O \times A)$ for the channel that delivers o and reports the action, and $K_o^0 : A \to \Delta(O \times \{*\})$ for the channel that delivers o and no record.

Lemma 17 (Fixed-channel and environment relevance). *Given Axioms* (A1)*,* (A5)*,* (A6)*,* (A7)*, and* (A8)*, the following equivalences hold.*

(a) *Axiom* (A3) *is equivalent to the existence of $o^+, o^- \in O$ such that*

$$(\delta_*, K_{o^+}) \succ (\delta_*, K_{o^-}).$$

(b) *Axiom* (A4) *is equivalent to the following condition: there is an outcome $o_* \in O$ such that, for every finite A with $|A| \geq 2$ and every full-support $q \in \Delta(A)$,*

$$(q, K_{o_*}^{\mathrm{id}}) \succ (q, K_{o_*}^0).$$

Proof. For part (a), apply action processing to either pure lottery in the two-action channel of Axiom (A3). Collapsing its support to a singleton gives K_{o^+} or K_{o^-}. Conversely, the singleton can be processed back to the corresponding action; the unrestricted row at the unreached action completes the pushforward to the original channel. Thus each pure lottery is indifferent to the corresponding singleton experiment. Weak order and block coherence transport the strict comparison in either direction.

For part (b), let o_* witness Axiom (A4) and first restrict its two lotteries to their supports. Action processing in both directions identifies the first with fair binary full revelation and the second with a fair binary channel whose two rows are the same. Record processing in both directions identifies the latter channel with silence. Hence the fixed-channel axiom is equivalent to

$$((\tfrac{1}{2}, \tfrac{1}{2}), K_{o_*}^{\mathrm{id}}) \succ ((\tfrac{1}{2}, \tfrac{1}{2}), K_{o_*}^0) \tag{6}$$

on a binary action set.

Now let p have full support on a binary set. Choose $0 < \varepsilon \leq 2 \min_i p_i$ and an interim record y with $P(y \mid i) = \varepsilon/(2p_i)$. The branch has probability ε and posterior $(\frac{1}{2}, \frac{1}{2})$. Compare two continuation plans: after that branch the first reveals the action and the second is silent; after every other branch both are silent. Equation (6) and branchwise continuation give a strict comparison of the two compounds. Full revelation can be garbled into the first compound, and the second compound can be garbled into silence. Record processing and transitivity therefore give $(p, K_{o_*}^{\mathrm{id}}) \succ (p, K_{o_*}^0)$.

Finally, let q have full support on any finite A with $|A| \geq 2$, and partition A into two nonempty cells. The channel that reports only the cell is, by action processing in both directions, indifferent to binary full revelation under the induced binary prior; silence on A is similarly indifferent to binary silence. The cell-reporting channel is therefore strictly preferred to silence, and full revelation is weakly preferred to the cell report by record processing. This proves the displayed comparison for every q and A at o_*. The converse follows by taking a fair binary prior and reversing the support identifications used above. □

Whenever Axiom (A4) is imposed below, fix one of its witnessing outcomes and denote it by o_*. Every channel between nonempty finite sets $P : A \to \Delta(X)$ has the constant-payoff lift

$$K_P(o, x \mid a) := \mathbf{1}_{\{o = o_*\}} P(x \mid a). \tag{7}$$

Throughout this appendix, comparisons of pure record channels mean comparisons of these lifts in the model already defined in the main text:

$$q \succeq_P q' \quad :\Longleftrightarrow \quad q \succeq_{K_P} q'.$$

We similarly write $(q,P) \succeq (r,Q)$ for the main-text two-block comparison of (q,K_P) and (r,K_Q). Likewise, $P \sqcup Q$, block-supported lotteries, action and record processing, and a compound $P_1 \triangleright \{Q^x\}$ are shorthand for the corresponding main-text constructions after applying (7). The lift commutes with each construction up to canonical tagged or singleton-record bijections; those bijections are neutral by Axiom (A6) applied in both directions. No new primitive relation or behavioural condition is introduced here.

Two pieces of notation will be used repeatedly. If $S : A \to \Delta(B)$ is an action processor, let $S^q P : B \to \Delta(X)$ denote any Bayesian pushforward satisfying

$$(qS)(b)(S^q P)(x \mid b) = \sum_a q(a)S(b \mid a)P(x \mid a).$$

Its rows at zero-mass reports are arbitrary, exactly as in (1). For channels $P : A \to \Delta(X)$ and $Q : B \to \Delta(Y)$, their independent product is

$$(P \otimes Q)((x,y) \mid (a,b)) := P(x \mid a)Q(y \mid b).$$

For $q \in \Delta(A)$ and $P : A \to \Delta(X)$, write

$$m_{qP}(x) := \sum_a q(a)P(x \mid a), \qquad r_x^{qP}(a) := \frac{q(a)P(x \mid a)}{m_{qP}(x)} \quad \text{when } m_{qP}(x) > 0,$$

and define the finite posterior law

$$\mu_{qP} := \sum_{x : m_{qP}(x) > 0} m_{qP}(x)\, \delta_{r_x^{qP}}.$$

Let Id_A be full revelation and U_A the one-record uninformative channel. With logarithms in the fixed base used in the main text, put

$$I_{qP}(A;X) := H(m_{qP}) - \sum_a q(a)H(P(\cdot \mid a)) = H(q) - \int_{\Delta(A)} H(r)\, d\mu_{qP}(r).$$

Proposition 18 (Pure-trace lemma)**.** *Suppose the preference family satisfies main-text Axioms* (A1)*,* (A2)*, and* (A4)–(A8)*. Then, for every record channel between nonempty finite sets* $P : A \to \Delta(X)$ *and every* $q, q' \in \Delta(A)$*,*

$$q \succeq_P q' \quad \Longleftrightarrow \quad I_{qP}(A;X) \geq I_{q'P}(A;X).$$

The same numerical scale represents every block-supported comparison: for every nonempty finite family $\{P_i : A_i \to \Delta(X_i)\}_{i \in J}$ *of channels between nonempty finite sets, distinct* $i,j \in J$*, and priors* $q_i \in \Delta(A_i)$ *and* $q_j \in \Delta(A_j)$*,*

$$q_i^i \succeq_{\bigsqcup_{k \in J} P_k} q_j^j \quad \Longleftrightarrow \quad I_{q_i P_i}(A_i;X_i) \geq I_{q_j P_j}(A_j;X_j).$$

A.1 Proof

Assume the hypotheses of Proposition 18. Every appeal below to weak order, continuity, block coherence, record processing, action processing, branchwise continuation, or positive trace orientation is an appeal to Axiom (A1), (A2), (A5), (A6), (A7), (A8), or (A4), respectively, on the constant-payoff lifts. Thus the proof has no auxiliary axiom system.

How to read the proof. The argument has five interfaces. Each stage ends with exactly the object needed by the next one:

Stage 1: replace channels by Bayes-plausible posterior laws, derive public-mixture independence, and cardinalise each fixed-prior quotient using only the segment closedness supplied by primitive continuity;

Stage 2: make relabelling, undetectable distinctions, and restriction to the positive support exact consequences of action processing;

Stage 3: transport those affine values through independent products and into one cross-prior block scale, allowing provisionally a bilinear interaction;

Stage 4: identify reached-branch coefficients, prove their cocycle, and align the scales on nested support faces;

Stage 5: compare product and sequential decompositions, eliminate the interaction, and invoke Faddeev's recursion to obtain Shannon entropy and hence mutual information.

The product-coefficient and branch tangent-space calculations are the two technical subarguments. A reader interested in the logical spine may use their lemma statements as interfaces. Product calibration is placed before branch calibration because this makes the scalar algebra easier to read. The early full-revelation normalisation and exact relabelling lemma below supply the coherence needed by that order of presentation.

A.1.1 Stage 1: posterior-law quotient and affine fibre values

For a finite action set A, let X_A be the set of finitely supported probability measures on $\Delta(A)$, and let

$$b(\mu) := \int_{\Delta(A)} s\, d\mu(s), \qquad \mathcal{M}_q := \{\mu \in \mathsf{X}_A : b(\mu) = q\}.$$

Thus $\mathcal{M}_q$ is the mixture set of finite Bayes-plausible posterior laws with prior q, and $\mu_{qP} \in \mathcal{M}_q$ for every finite channel P.

Lemma 19 (Coherent comparisons across blocks)**.** *The relation on prior–channel pairs defined above is complete and transitive. It is unaffected by adding or deleting blocks other than the two being compared. In particular, for fixed q the induced relation on channels is a weak order, and*

$$q \succeq_P r \quad\Longleftrightarrow\quad (q, P) \succeq (r, P).$$

For fixed alphabets, its graph is closed under entrywise convergence of the two channels and convergence of the two priors.

Proof. Completeness is Axiom (A1) in the two-block environment. To prove transitivity, suppose $(q_1, P_1) \succeq (q_2, P_2)$ and $(q_2, P_2) \succeq (q_3, P_3)$. Put the three pairs in the common environment $P_1 \sqcup P_2 \sqcup P_3$. Axiom (A5) transports the two hypotheses to that environment, Axiom (A1) composes them there, and Axiom (A5) deletes the middle block. Applying Axiom (A5) with the ordered pair of block labels reversed gives

$$(q_2, P_2) \succeq (q_1, P_1) \quad\Longleftrightarrow\quad q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0,$$

so reversal is also canonical. The displayed within-channel equivalence is the duplication clause of Axiom (A5). Finally, a block channel is a continuous affine function of its component channels. Closedness on fixed alphabets is therefore exactly Axiom (A2) applied to the common block environment. □

Lemma 20 (Posterior-law quotient)**.** *Fix a full-support $q \in \Delta(A)$. If finite channels $P : A \to \Delta(X)$ and $Q : A \to \Delta(Y)$ satisfy*

$$\mu_{qP} = \mu_{qQ},$$

then they are mutually obtainable by record post-processing. Consequently

$$(q, P) \sim (q, Q),$$

and either channel may be substituted for the other in every block comparison at prior q. Hence the fixed-q order of channels descends to a well-defined weak order $\succeq_q$ on $\mathcal{M}_q$.

Proof. We give the finite matching argument because it also handles repeated and zero-probability records. Put

$$m_P(x) := \sum_a q(a)P(x \mid a), \qquad m_Q(y) := \sum_a q(a)Q(y \mid a).$$

For every posterior s occurring with positive probability, let

$$X_s := \{x : m_P(x) > 0,\ r_x^{qP} = s\}, \qquad Y_s := \{y : m_Q(y) > 0,\ r_y^{qQ} = s\}.$$

Equality of posterior laws gives the common positive mass

$$\lambda_s := \sum_{x \in X_s} m_P(x) = \sum_{y \in Y_s} m_Q(y).$$

For $x \in X_s$, define a stochastic record kernel by

$$T(y \mid x) := \begin{cases} m_Q(y)/\lambda_s, & y \in Y_s, \\ 0, & y \notin Y_s. \end{cases}$$

Choose arbitrary stochastic rows at records x with $m_P(x) = 0$. Full support of q implies that such a record has $P(x \mid a) = 0$ for every a.

If $y \in Y_s$, Bayes' rule gives

$$\sum_{x \in X_s} P(x \mid a) = \frac{\lambda_s s(a)}{q(a)}, \qquad Q(y \mid a) = \frac{m_Q(y)s(a)}{q(a)}.$$

It follows that

$$(PT)(y \mid a) = \sum_{x \in X_s} P(x \mid a)\frac{m_Q(y)}{\lambda_s} = Q(y \mid a).$$

Both sides vanish when $m_Q(y) = 0$, again by full support. Thus $PT = Q$. Interchanging P and Q constructs T' with $QT' = P$.

Axiom (A6) applied to T and T' yields the two weak comparisons; Lemma 19 puts the reverse comparison into the same block order and gives indifference. If P, P' have the same posterior law, place $(q, P), (q, P')$ and the two objects being compared in one finite block environment. Transitivity then permits replacement of P by P' on either side. The quotient order is consequently independent of the implementing channel and inherits completeness and transitivity from Lemma 19. □

Every element of $\mathcal{M}_q$ is implementable when q has full support. Indeed, if

$$\mu = \sum_{k=1}^{n} \lambda_k \delta_{s_k}, \qquad \lambda_k > 0, \qquad \sum_k \lambda_k s_k = q,$$

then

$$C_\mu(k \mid a) := \frac{\lambda_k s_k(a)}{q(a)} \tag{8}$$

defines a channel and $\mu_{qC_\mu} = \mu$.

For channels $P : A \to \Delta(X)$ and $R : A \to \Delta(Z)$ and $t \in [0,1]$, let $tP \oplus (1-t)R$ be the publicly tagged mixture: it reports which component was selected as well as that component's record. Its posterior law is

$$\mu_{q,tP\oplus(1-t)R} = t\mu_{qP} + (1-t)\mu_{qR}. \tag{9}$$

Lemma 21 (Public-mixture independence)**.** *Let q have full support. For all finite channels P, Q, R on the action set A and every $t \in (0,1)$,*

$$(q,P) \succeq (q,Q) \quad \Longleftrightarrow \quad (q, tP \oplus (1-t)R) \succeq (q, tQ \oplus (1-t)R).$$

Equivalently, the quotient order on $\mathcal{M}_q$ satisfies mixture independence.

Proof. If necessary, embed the record alphabets of P, Q, R into one tagged disjoint union and fill the unused coordinates with zeros. This does not change any posterior law, so Lemma 20 makes the padding neutral.

Let $C : A \to \Delta(\{0,1\})$ be the uninformative first-stage channel with $C(0 \mid a) = t$ for every a. Both branches are reached, and their posterior is q. The compounds

$$C \triangleright \{P, R\}, \qquad C \triangleright \{Q, R\}$$

have the posterior laws on the right-hand side of (9); they differ from the displayed public mixtures only by record labels. If $(q,P) \succeq (q,Q)$, Axiom (A8) applied to branch 0, together with reflexivity on the common branch 1, gives the aggregate comparison.

Conversely, suppose the aggregate comparison holds but $(q,P) \not\succeq (q,Q)$. Completeness gives $(q,Q) \succ (q,P)$. Apply the strict clause of Axiom (A8) with the two continuation profiles reversed. Branch 0 is reached with probability $t > 0$, so it yields the strict reverse aggregate comparison, contradicting the assumed weak one. Posterior-law substitution transfers both conclusions between the compounds and the tagged mixtures. $\square$

Lemma 22 (Affine posterior-law representation)**.** *For every full-support $q \in \Delta(A)$ there is an affine functional*

$$F_q : \mathcal{M}_q \longrightarrow \mathbb{R}$$

such that, for all finite channels P, Q on A,

$$(q,P) \succeq (q,Q) \quad \Longleftrightarrow \quad F_q(\mu_{qP}) \geq F_q(\mu_{qQ}). \tag{10}$$

If $|A| \geq 2$, F_q is unique up to a positive affine transformation. If $|A| = 1$, $\mathcal{M}_q$ is a singleton and we take $F_q \equiv 0$.

Proof. By Lemma 20 and (8), $\succeq_q$ is a weak order on the mixture set $\mathcal{M}_q$. Lemma 21 gives mixture independence. It remains only to verify Herstein–Milnor's segment-continuity condition. This is where primitive Axiom (A2) is used; no topology on the collection of all finite posterior laws is needed.

Fix $\mu, \nu, \rho \in \mathcal{M}_q$ and choose finite implementing channels P, Q, R. For $t \in [0,1]$ define

$$M_t := tP \oplus (1-t)Q.$$

All M_t have the single fixed record alphabet given by the tagged union of the record alphabets of P and Q. If $t_n \to t$, then $M_{t_n} \to M_t$ entrywise. After adjoining the fixed block R, all comparisons occur in one fixed finite action and record alphabet. Axiom (A2) therefore makes both sets

$$\{t \in [0,1] : M_t \succeq_q R\}, \qquad \{t \in [0,1] : R \succeq_q M_t\}$$

closed. By (9), these are exactly

$$\{t : t\mu + (1-t)\nu \succeq_q \rho\}, \qquad \{t : \rho \succeq_q t\mu + (1-t)\nu\}.$$

Thus the Herstein–Milnor mixture-space theorem [30] applies and supplies an affine representation F_q. When $|A| \geq 2$, Axiom (A4) makes the order nonconstant, so the usual positive-affine uniqueness conclusion applies. For a singleton action set, Bayes plausibility forces every posterior law to equal δ_q. $\square$

The preceding proof uses only closedness along each particular public-mixture segment. It neither assumes nor proves closedness under arbitrary weakly convergent sequences whose support sizes grow. It also requires no pointwise integrand on $\Delta(A)$: every subsequent calculation uses only the affinity of F_q on $\mathcal{M}_q$ and its linear part on the affine hull.

Let

$$\delta_q \in \mathcal{M}_q$$

be the posterior law of the uninformative channel and let

$$\chi_q := \sum_{a\in A} q(a)\delta_{e_a} \in \mathcal{M}_q$$

be the posterior law of full revelation.

Lemma 23 (Canonical fibre normalization)**.** *The representatives in Lemma 22 may be chosen so that*

$$F_q(\delta_q) = 0, \qquad F_q(\mu) \geq 0 \quad (\mu \in \mathcal{M}_q), \tag{11}$$

and, whenever $|A| \geq 2$,

$$F_q(\chi_q) = 1. \tag{12}$$

These two anchors select a unique affine representative on every nondegenerate full-support fibre. On a singleton fibre we retain $F_q \equiv 0$.

Proof. Every channel P can be record-postprocessed to the uninformative channel by deleting its record. Axiom (A6) and (10) therefore imply

$$F_q(\mu_{qP}) \geq F_q(\delta_q).$$

Every law in $\mathcal{M}_q$ is implemented by (8), so δ_q is a minimum of F_q on the whole fibre. Subtract $F_q(\delta_q)$ to obtain (11).

If $|A| \geq 2$, Axiom (A4) says that full revelation is strictly preferred to no information at every full-support prior. Hence $F_q(\chi_q) > 0$. Dividing by this positive number gives (12) without changing any comparison. If two affine representatives have both anchor values, positive-affine uniqueness forces the additive constant to be zero and the positive multiplier to be one. The singleton convention is immediate. $\square$

A.1.2 Stage 2: transport across priors and support faces

The remaining front-end issue is transport. The same argument handles renaming actions, deleting distinctions that the channel cannot detect, and passing to the positive support of a boundary prior. Keeping these facts in one lemma makes clear that all three are consequences of action processing, not extra invariance conventions.

For a bijection $\pi : A \to A'$, write $q\pi$ for the transported prior and $\pi_{\#}\mu$ for the posterior law obtained by transporting every atom of μ. If $B = \mathrm{supp}(q)$, write q_B for the full-support restriction of q to B, P_B for the restricted channel, and $\pi_B\mu$ for the law obtained by restricting every posterior atom to B.

Lemma 24 (Relabelling, undetectable distinctions, and support transport)**.** *The canonically normalized fibre values have the following properties.*

(a) Exact relabelling. *If q has full support and $\pi : A \to A'$ is a bijection, then*

$$F_{q\pi}^{A'}(\pi_{\#}\mu) = F_q^A(\mu) \qquad (\mu \in \mathcal{M}_q). \tag{13}$$

(b) Undetectable distinctions. *Let $S : A \to B$ be an onto deterministic map, let q have full support, and suppose that $P : A \to \Delta(X)$ is constant on every fibre of S. Then*

$$(q, P) \sim (qS, S^q P). \tag{14}$$

In particular, if G_S reports only $S(a)$, then $(q, G_S) \sim (qS, \mathrm{Id}_B)$.

(c) Support restriction. *For an arbitrary prior $q \in \Delta(A)$ with nonempty support $B := \mathrm{supp}(q)$ and every channel P,*

$$(q, P) \sim (q_B, P_B). \tag{15}$$

Consequently comparisons at q depend only on the rows indexed by B, and

$$(q, P) \succeq (q, Q) \quad \iff \quad (q_B, P_B) \succeq (q_B, Q_B). \tag{16}$$

Every atom of a law $\mu \in \mathcal{M}_q$ is supported on B, and the definition

$$F_q^A(\mu) := F_{q_B}^B(\pi_B \mu) \tag{17}$$

represents all continuation comparisons at the boundary prior q.

Proof. Part (a). Regard π as a deterministic action processor. Axiom (A7) gives

$$(q, P) \succeq (q\pi, P\pi^{-1}),$$

where $(P\pi^{-1})(x \mid a') = P(x \mid \pi^{-1}(a'))$. Applying the same axiom to π^{-1} gives the reverse comparison, so the two pairs are indifferent. Apply this simultaneously to channels implementing arbitrary $\mu, \nu \in \mathcal{M}_q$ and use Lemma 19 in a common four-block environment. One obtains

$$F_q^A(\mu) \geq F_q^A(\nu) \quad \iff \quad F_{q\pi}^{A'}(\pi_{\#}\mu) \geq F_{q\pi}^{A'}(\pi_{\#}\nu).$$

Thus $G(\mu) := F_{q\pi}^{A'}(\pi_{\#}\mu)$ is an affine representation of the same nonconstant order as F_q^A. Relabelling sends δ_q to $\delta_{q\pi}$ and χ_q to $\chi_{q\pi}$. The two anchor equalities in Lemma 23, followed by affine uniqueness, give $G = F_q^A$. If A is a singleton, both sides vanish.

Part (b). Since q has full support and S is onto, qS has full support. One direction of (14) is Axiom (A7). For the reverse direction, define the Bayesian refinement kernel

$$R(a \mid b) := \begin{cases} q(a)/(qS)(b), & S(a) = b, \\ 0, & S(a) \neq b. \end{cases}$$

Then $(qS)R = q$. Since P is constant on each fibre of S, the pushforward $S^q P$ has at b precisely the common row of P on $S^{-1}(b)$, and pushing it back through R recovers P. Axiom (A7) applied to $(qS, S^q P)$ and R therefore gives the reverse weak comparison. For the last assertion, G_S is S-measurable and its pushforward is Id_B.

Part (c). Choose $b_0 \in B$ and let $S : A \to B$ fix every $b \in B$ and send every $a \notin B$ to b_0. Since q assigns zero mass outside B,

$$qS = q_B, \qquad S^q P = P_B.$$

Axiom (A7) gives $(q, P) \succeq (q_B, P_B)$. Conversely, let $\iota : B \hookrightarrow A$ be the inclusion. Then $q_B\iota = q$. The exact Bayesian completion equation pins down the processed channel on B and leaves all zero-mass rows in $A \setminus B$ arbitrary. The all-completions clause of Axiom (A7) therefore permits us to choose the completion to be the original P, yielding $(q_B, P_B) \succeq (q, P)$. This proves (15); applying it to P and Q in a common block environment proves (16). It also shows immediately that channels agreeing on B are indifferent at q.

Finally, if $\mu = \sum_k \lambda_k \delta_{s_k} \in \mathcal{M}_q$ with $\lambda_k > 0$, then for $a \notin B$,

$$0 = q(a) = \sum_k \lambda_k s_k(a).$$

Nonnegativity forces $s_k(a) = 0$ for every k. Thus every atom lies on the face $\Delta(B)$ and $\pi_B\mu \in \mathcal{M}_{q_B}$. The comparison equivalence just proved and the full-support representation on B show that (17) represents precisely the boundary continuation order. This definition is also compatible with relabelling by part (a). □

At this point every later calculation may work entirely with the affine functionals F_q, their linear parts on the affine hulls of $\mathcal{M}_q$, and the exact transport identities above. No global posterior-law continuity statement and no posterior-separable integrand are needed. From now on ambient action-set superscripts are suppressed unless two action sets occur in the same formula; a boundary-prior subscript always means the representative transported from its positive support. The next two stages derive product and branch scale coherence, with Axiom (A8) providing the essential cardinal alignment across reached branches.

A.1.3 Stage 3: products and a common comparison scale

The value F_q compares experiments only within the fibre of one prior. Embedding experiments from two priors into their product fibre will put both values on one cardinal comparison scale. We first derive the only separability fact about independent products needed for that construction. Write $P \succeq_q Q$ for $(q, P) \succeq (q, Q)$, and similarly for strict preference. For a channel $P : A \to \Delta(X)$, let $\widetilde{P}$ denote its lift to $A \times B$, constant in the B-coordinate; for $R : B \to \Delta(Y)$, let $\widetilde{R}$ denote the symmetric lift, constant in the A-coordinate. Posterior laws may be substituted throughout by Lemma 20; in particular, adding null records and permuting record labels has no effect on a comparison.

Lemma 25 (Cancellation of a common independent background)**.** *Let $p \in \Delta(A)$ and $r \in \Delta(B)$ have full support. For channels P, Q on A and a channel R on B,*

$$P \otimes R \succeq_{p\otimes r} Q \otimes R \quad \Longleftrightarrow \quad P \succeq_p Q. \tag{BI}$$

The symmetric assertion holds with the two coordinates interchanged. The equivalence, including its strict form, remains valid when a posterior reached in the background coordinate lies on a proper support face.

Proof. Pad P and Q by null records so that they have the same record alphabet. Regard the lift $\widetilde{R}$ of R as a first-stage experiment on $A \times B$, followed on every reached branch by $\widetilde{P}$, respectively $\widetilde{Q}$. Up to the harmless permutation of the two record coordinates, the resulting compounds are $P \otimes R$ and $Q \otimes R$.

If y is reached under R, its posterior on $A \times B$ is

$$p \otimes r_y, \qquad r_y := r_y^{r,R}.$$

The coordinate projection $A \times \operatorname{supp}(r_y) \to A$ and the support-transport clauses of Lemmas 24 and 24 give

$$(p \otimes r_y, \widetilde{P}) \sim (p, P), \qquad (p \otimes r_y, \widetilde{Q}) \sim (p, Q). \tag{BI$'$}$$

Thus every reached branch has the same weak comparison as P versus Q at p. If $P \succeq_p Q$, the weak clause of Axiom (A8) applied to these branches proves the forward implication in (BI).

For the converse, suppose that $P \not\succeq_p Q$. Completeness gives $Q \succ_p P$. By (BI′), the reversed comparison is strict on every reached branch; at least one such branch has positive mass. The strict clause of Axiom (A8) therefore gives $Q \otimes R \succ_{p\otimes r} P \otimes R$, contradicting $P \otimes R \succeq_{p\otimes r} Q \otimes R$. This proves the converse and also records why weak branch monotonicity alone would not suffice. Interchange the coordinates for the symmetric assertion. ☐

For finite posterior laws define

$$\left(\sum_i m_i \delta_{u_i}\right) \otimes \left(\sum_j n_j \delta_{v_j}\right) := \sum_{i,j} m_i n_j \delta_{u_i \otimes v_j}.$$

Bayes' rule gives

$$\mu_{p\otimes r, P\otimes R} = \mu_{pP} \otimes \mu_{rR}. \tag{PL}$$

Lemma 26 (Product gauge)**.** *There are positive rescalings $G_q = c_q F_q$ of the zero-normalised affine representatives and a single number $\kappa \in \mathbb{R}$ such that*

$$G_{p\otimes r}(\mu \otimes \nu) = G_p(\mu) + G_r(\nu) + \kappa G_p(\mu) G_r(\nu) \tag{QA}$$

for all full-support priors p, r and all $\mu \in \mathcal{M}_p$, $\nu \in \mathcal{M}_r$. These rescalings can be chosen exactly invariant under bijections. If a factor is a singleton, its representative is zero and the same formula holds. Finally,

$$1 + \kappa G_r(\nu) > 0 \tag{POS}$$

whenever the r-fibre is nonconstant.

Proof. We give the algebra because both the common value of κ and the strict inequality (POS) are used later. Initially retain the notation F_q. For nondegenerate p, r, put

$$L_{p,r}(\mu, \nu) := F_{p\otimes r}(\mu \otimes \nu), \qquad x := F_p(\mu), \quad y := F_r(\nu).$$

The map $L_{p,r}$ is separately affine. By Lemma 25, every first-coordinate slice represents the F_p-order and every second-coordinate slice represents the F_r-order. Affine-utility uniqueness, first on a slice and then as the other coordinate varies affinely, therefore gives unique constants $A_{p,r}, B_{p,r} > 0$ and $C_{p,r} \in \mathbb{R}$ such that

$$L_{p,r}(\mu, \nu) = A_{p,r} x + B_{p,r} y + C_{p,r} xy. \tag{18}$$

Here is the coefficient argument. Fixing ν first gives $L_{p,r} = \alpha(\nu) x + \gamma(\nu)$ with $\alpha(\nu) > 0$. At $\mu = \delta_p$, the function $\gamma(\nu) = L_{p,r}(\delta_p, \nu)$ is an affine representative of the F_r-order and vanishes at δ_r; hence $\gamma(\nu) = B_{p,r} y$ for some $B_{p,r} > 0$. Choose $\bar\mu$ with $\bar x := F_p(\bar\mu) \neq 0$. The other slice has the form

$$L_{p,r}(\bar\mu, \nu) = A_{p,r} \bar x + D_{p,r} y,$$

where $A_{p,r} > 0$ is its value slope at $\nu = \delta_r$. Comparing this with $\alpha(\nu)\bar x + B_{p,r} y$ shows that $\alpha(\nu) = A_{p,r} + C_{p,r} y$, where $C_{p,r} := (D_{p,r} - B_{p,r})/\bar x$. This proves (18); the symmetric calculation shows that the two slice slopes obey

$$A_{p,r} + C_{p,r} y > 0, \qquad B_{p,r} + C_{p,r} x > 0 \tag{19}$$

throughout the two value ranges.

Exact relabelling makes the evaluations under the two associations of a triple product equal. Comparing the coefficients of x, y, z gives

$$A_{p\otimes r,s}A_{p,r} = A_{p,r\otimes s}, \tag{20}$$
$$A_{p\otimes r,s}B_{p,r} = B_{p,r\otimes s}A_{r,s}, \tag{21}$$
$$B_{p\otimes r,s} = B_{p,r\otimes s}B_{r,s}. \tag{22}$$

The coordinate swap similarly gives

$$B_{r,p} = A_{p,r}, \qquad A_{r,p} = B_{p,r}, \qquad C_{r,p} = C_{p,r}. \tag{SW}$$

This coefficient comparison is legitimate: the laws $(1-t)\delta_p + t\chi_p$ make x range over a nondegenerate interval, and the same holds for y and z. Let $\rho(p,r) := B_{p,r}/A_{p,r}$. Dividing (21) by (20) yields

$$\rho(p,r) = \rho(p, r\otimes s)A_{r,s}.$$

Fix a nondegenerate reference prior q_0 and set $g(r) := \rho(q_0, r)$. Then

$$A_{r,s} = \frac{g(r)}{g(r\otimes s)}. \tag{23}$$

The uniqueness of the coefficients in (18), applied after product relabellings, shows that g is invariant under bijections. Consequently the positive rescaling $G_p := g(p)F_p$ preserves exact relabelling. Equation (23) makes the new first coefficient equal to one, and (SW) makes the second coefficient equal to one as well. We henceforth use G for these gauged representatives.

Write the remaining coefficient as $\kappa_{p,r}$. Associativity of

$$x \oplus_{p,r} y := x + y + \kappa_{p,r}xy$$

and comparison of the coefficients of xy, xz, yz give

$$\kappa_{p,r} = \kappa_{p,r\otimes s}, \qquad \kappa_{p\otimes r,s} = \kappa_{p,r\otimes s}, \qquad \kappa_{p\otimes r,s} = \kappa_{r,s}. \tag{KA}$$

(The xyz coefficient gives the product of the corresponding two sides and adds no restriction.) Taking $s = q_0$ in (KA) shows that $\kappa_{p,r} = \kappa_{r,q_0}$, so it is independent of p; the swap in (SW) makes it symmetric, hence independent of r. Call the common value κ. After the gauge, (19) is exactly $1 + \kappa G_r(\nu) > 0$ and its symmetric counterpart.

For a singleton prior set $G \equiv 0$; the canonical bijections $A \cong A \times \{*\}$ give (QA). For a boundary prior, define G_r by exact transport from its positive-probability support, as in Lemma 24. This convention is relabelling invariant. □

Corollary 27 (Exact relabelling after the product gauge)**.** *For every bijection $\pi : A \to A'$, prior $q \in \Delta(A)$, and $\mu \in \mathcal{M}_q$,*

$$G^{A'}_{q\pi}(\pi_{\#}\mu) = G^A_q(\mu).$$

Proof. The initial equality is Lemma 24; the proof of Lemma 26 shows that its gauge factor is invariant under bijections. □

Lemma 28 (Cross-prior block representation)**.** *For arbitrary finite priors $q \in \Delta(A)$ and $r \in \Delta(B)$ and channels P, Q on the respective action sets,*

$$q^0 \succeq_{P\sqcup Q} r^1 \iff G_q(\mu_{qP}) \geq G_r(\mu_{r,Q}). \tag{BC}$$

Proof. First suppose that q, r have full support. By (QA) and $G_q(\delta_q) = G_r(\delta_r) = 0$,

$$G_{q\otimes r}(\mu_{qP} \otimes \delta_r) = G_q(\mu_{qP}), \qquad G_{q\otimes r}(\delta_q \otimes \mu_{r,Q}) = G_r(\mu_{r,Q}).$$

The two laws on the left lie in the same fibre and are implemented by $P \otimes U_B$ and $U_A \otimes Q$, so the affine fibre representation compares them by the displayed numbers.

It remains only to remove the uninformative product coordinates. Projection $A \times B \to A$ and Axiom (A7) give $(q \otimes r, P \otimes U_B) \succeq (q, P)$. Conversely the stochastic embedding

$$T((a, b) \mid a') = \mathbf{1}_{\{a=a'\}} r(b)$$

sends q to $q \otimes r$ and its Bayesian pushforward sends P to $P \otimes U_B$; Axiom (A7) gives the reverse comparison. Thus the two pairs are indifferent. The same argument identifies $(q \otimes r, U_A \otimes Q)$ with (r, Q). Block coherence and transitivity now prove (BC). Arbitrary priors follow by restricting both pairs and both representatives to their supports; singleton supports use the zero convention. The singleton record tags created by these projections and embeddings are the canonical relabellings covered by the convention at the start of the appendix. □

A.1.4 Stage 4: sequential decomposition and coherent face scales

Changing a continuation on one reached branch changes the posterior law by a signed measure with zero total mass and zero barycentre. The following space is exactly the space of such feasible directions. For a nonempty finite set A, let

$$W_A := \left\{ \eta : \begin{array}{l} \eta \text{ is a finitely supported signed measure on } \Delta(A), \\ \eta(\Delta(A)) = 0, \qquad \int p \, d\eta(p) = 0 \end{array} \right\}.$$

If $\iota : B \hookrightarrow A$ is an injection, let $i_\iota : W_B \to W_A$ push every posterior forward along ι; for a literal subset $B \subseteq A$, write i_B for the inclusion case. Let L^A_q be the linear part of the affine functional G_q on the affine hull of $\mathcal{M}_q$.

Lemma 29 (Branch aggregation)**.** *Let q have full support on A. For every reached posterior r with $B := \operatorname{supp}(r)$ and $|B| \geq 2$, there is a unique $\beta_A(q, r) > 0$ satisfying*

$$L^A_q \circ i_B = \beta_A(q, r) L^B_{r_B} \quad \text{on } W_B. \tag{24}$$

It depends only on q, r and the face inclusion, not on the experiment that reaches r. If a first-stage channel P_1 has reached branches of masses m_x and posteriors r_x, then every continuation profile satisfies

$$G_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = G_q(\mu_{qP_1}) + \sum_{x : m_x > 0} m_x \beta_A(q, r_x) G_{r_x}(\mu_{r_x, Q^x}). \tag{BA}$$

For singleton r_x, the continuation value is zero and the corresponding coefficient may be fixed by any positive convention.

Proof. We first record the common tangent space. If q has full support, then

$$\operatorname{span}\{\mu - \nu : \mu, \nu \in \mathcal{M}_q\} = W_A. \tag{25}$$

Only the reverse inclusion needs proof. For $0 \neq \eta \in W_A$, write its Jordan decomposition as $\eta = \eta^+ - \eta^-$, where both positive parts have the same mass $c > 0$ and, after division by c, the same barycentre b. Choose $\varepsilon > 0$ small enough that $\tau = (q - \varepsilon b)/(1 - \varepsilon)$ belongs to $\Delta(A)$. Then

$$\mu^\pm := \frac{\varepsilon}{c}\eta^\pm + (1 - \varepsilon)\delta_\tau \in \mathcal{M}_q, \qquad \eta = \frac{c}{\varepsilon}(\mu^+ - \mu^-),$$

which proves (25). The same argument intrinsically on a face gives the space W_B.

Fix a nondegenerate r supported on B. It is reachable from q: choose

$$0 < \varepsilon < \min\left\{1, \min_{b \in B} \frac{q(b)}{r(b)}\right\}$$

and use a two-record experiment whose first record has conditional probability $\varepsilon r(a)/q(a)$. That record has mass ε and posterior r.

Implement $\nu, \nu' \in \mathcal{M}_{r_B}$ by channels on B, extend their off-support rows arbitrarily to A, and pad their record alphabets if necessary. Vary only the continuation on the branch reaching r from ν to ν'. The aggregate posterior law changes by $\varepsilon i_B(\nu - \nu')$. Axiom (A8), support transport, and the strict clause give all three signs:

$$\begin{aligned}
L^B_{r_B}(\nu - \nu') > 0 &\Longrightarrow L^A_q i_B(\nu - \nu') > 0,\\
L^B_{r_B}(\nu - \nu') < 0 &\Longrightarrow L^A_q i_B(\nu - \nu') < 0,\\
L^B_{r_B}(\nu - \nu') = 0 &\Longrightarrow L^A_q i_B(\nu - \nu') = 0.
\end{aligned}$$

The second line is the first applied after swapping the continuations. For the third, indifference supplies both weak branch comparisons, and the weak part of Axiom (A8) supplies both aggregate inequalities. By (25), the sign agreement extends by positive scaling to all of W_B. Positive trace orientation makes

$$L^B_{r_B}(\chi_{r_B} - \delta_{r_B}) = G_{r_B}(\chi_{r_B}) > 0,$$

so $L^B_{r_B} \neq 0$. Two nonzero linear functionals with the same positive half-space have the same kernel and are positive scalar multiples; this gives the unique coefficient in (24). Since both linear functionals in that equation are fixed before a reaching experiment is chosen, the coefficient is path-independent.

For the aggregation formula, put $B_x := \operatorname{supp}(r_x)$ and

$$\nu_x := \pi_{B_x} \mu_{r_x, Q^x} \in \mathcal{M}_{(r_x)_{B_x}}.$$

The compound law differs from the first-stage law by

$$\mu_{q, P_1 \triangleright \{Q^x\}} - \mu_{q P_1} = \sum_{x : m_x > 0} m_x \, i_{B_x}(\nu_x - \delta_{(r_x)_{B_x}}).$$

Apply L^A_q, use (24), and use $G_{r_x}(\delta_{r_x}) = 0$. Singleton summands vanish. This is (BA). $\square$

For an injection $\iota : B \hookrightarrow A$ and full-support $r \in \Delta(B)$, define

$$\beta_\iota(q, r) := \beta_A(q, \iota_\# r).$$

The image posterior $\iota_\# r$ is reachable by the two-record construction in the proof, and its support inclusion is precisely i_ι. Thus

$$L^A_q i_\iota = \beta_\iota(q, r) L^B_r \quad \text{on } W_B.$$

Lemma 30 (Cocycle, face coherence, and the chain gauge)**.** *The coefficients of Lemma 29 admit positive scales a^A_q which are invariant under relabelling and satisfy*

$$\beta_\iota(q, r) = \frac{a^A_q}{a^B_r} \tag{FS}$$

whenever $\iota : B \hookrightarrow A$, both sets are nondegenerate, q and r have full support on their respective sets, and $\beta_\iota(q, r)$ is the coefficient for reaching the face $\iota(B)$. Hence the representatives

$$V_q := \frac{G_q}{a_q}$$

satisfy the exact chain rule

$$V_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = V_q(\mu_{qP_1}) + \sum_{x:m_x>0} m_x V_{r_x}(\mu_{r_x,Q^x}). \tag{CR}$$

Singleton continuation terms are zero.

Proof. There are two normalization tasks: first solve the branch cocycle within each fixed action-set dimension, then remove the residual scalar attached to embedding an m-point face in an n-point simplex.

We first prove the cocycle before choosing scales. For nested injections $C \overset{\jmath}{\hookrightarrow} B \overset{\iota}{\hookrightarrow} A$ and full-support priors q, r, s on A, B, C, respectively, the defining identities on W_C are

$$L_q^A i_{\iota\circ\jmath} = \beta_\iota(q,r) L_r^B i_\jmath = \beta_\iota(q,r)\beta_\jmath(r,s) L_s^C.$$

The direct coefficient from q to s is unique because $L_s^C \neq 0$ by Axiom (A4). Therefore

$$\beta_{\iota\circ\jmath}(q,s) = \beta_\iota(q,r)\beta_\jmath(r,s). \tag{26}$$

This single identity covers full-support paths, full-to-face paths, and nested support faces.

For a fixed nondegenerate A, apply (26) to identity embeddings. Fix a full-support basepoint q_A and put

$$a_q^A := \beta_{\mathrm{id}_A}(q, q_A).$$

Then

$$\beta_{\mathrm{id}_A}(q,r) = \frac{a_q^A}{a_r^A}. \tag{WS}$$

Choosing q_A to be uniform and using exact relabelling of G makes these initial scales invariant under bijections. They remain free up to one positive multiplier for each cardinality.

For an injection $\iota : B \hookrightarrow A$, define its residual defect by

$$K(\iota) := \beta_\iota(q,r)\frac{a_r^B}{a_q^A}.$$

Equation (WS) shows that this number is independent of q and r: changing either prior merely inserts and cancels the corresponding within-set coefficient. Equation (26) gives

$$K(\iota \circ \jmath) = K(\iota)K(\jmath).$$

Exact relabelling implies that $K(\iota)$ depends only on $n = |A|$ and $m = |B|$; write it $K_{n,m}$. Thus

$$K_{n,\ell} = K_{n,m}K_{m,\ell}, \qquad K_{n,n} = 1.$$

Set $t_n := K_{n,2}$ and $t_2 := 1$. Then $K_{n,m} = t_n/t_m$. Replacing every a_q^A on an n-point set by $t_n a_q^A$ changes the defect to $K_{n,m}t_m/t_n = 1$, while preserving (WS) and relabel invariance. This proves (FS).

For a boundary prior, a_r means the scale on its support; the preceding construction says precisely that this agrees with the scale obtained by reaching that support as a face. Divide (BA) by a_q and use (FS). Every nondegenerate branch coefficient becomes

$$\frac{1}{a_q}\beta_A(q,r_x)G_{r_x} = \frac{1}{a_{r_x}}G_{r_x} = V_{r_x}.$$

Singleton values vanish, so their scales and coefficients may be fixed arbitrarily. This proves (CR). $\square$

A.1.5 Stage 5: compatibility and entropy identification

The product and sequential evaluations of full revelation now meet. Their compatibility first removes the provisional product interaction and then leaves a strongly additive uncertainty functional.

Lemma 31 (Interaction collapse and universal chain scale)**.** *The coefficient κ in (QA) is zero. Moreover there is a single $C > 0$ such that $a_q = C$ for every nondegenerate finite prior, read on its support. The same value may be assigned to singleton priors. Consequently the final representatives $V_q = G_q/C$ satisfy both*

$$V_{p\otimes r}(\mu \otimes \nu) = V_p(\mu) + V_r(\nu), \tag{PA}$$

$$V_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = V_q(\mu_{qP_1}) + \sum_{x:m_x>0} m_x V_{r_x}(\mu_{r_x,Q^x}). \tag{CR$'$}$$

The cross-prior bridge of Lemma 28 is valid with V in place of G.

Proof. In this proof only, put

$$H_G(q) := G_q(\chi_q), \qquad Z(q) := 1 + \kappa H_G(q).$$

For nondegenerate q, Axiom (A4) and zero normalisation give $H_G(q) > 0$, while the strict slice-slope (POS) gives

$$Z(q) > 0. \tag{27}$$

For a singleton, $H_G = 0$ and $Z = 1$.

Product revelation versus sequential revelation. For nondegenerate full-support p, r, product quasi-additivity gives

$$H_G(p \otimes r) = H_G(p) + H_G(r) + \kappa H_G(p) H_G(r). \tag{QH}$$

Compute the same full revelation by first revealing the A-coordinate and then the B-coordinate. The first-stage law is $\chi_p \otimes \delta_r$ and has value $H_G(p)$ by (QA). After action a is revealed, the posterior is $e_a \otimes r$ on the face $\{a\} \times B$; support restriction and relabelling identify the continuation value with $H_G(r)$. Every such branch has coefficient $a_{p\otimes r}/a_r$ by (FS). Since the branch weights sum to one, Lemma 29 gives

$$H_G(p \otimes r) = H_G(p) + \frac{a_{p\otimes r}}{a_r} H_G(r). \tag{SA}$$

Comparing (QH) and (SA), and using $H_G(r) > 0$, yields

$$\frac{a_{p\otimes r}}{a_r} = Z(p). \tag{R1}$$

Revealing B first gives symmetrically

$$\frac{a_{p\otimes r}}{a_p} = Z(r). \tag{R2}$$

Thus $a_r Z(p) = a_p Z(r)$ for every pair of nondegenerate priors, even on different finite sets. By (27), there is a universal $C > 0$ such that

$$a_q = C Z(q). \tag{AS}$$

Assign $a = C$ on singleton supports.

If $\kappa = 0$, then $Z \equiv 1$ and the conclusion already follows. It remains to rule out $\kappa \neq 0$.

The grouping weight. Let q have full support and let $\mathcal{P} = \{A_1, \dots, A_m\}$ partition its action set into nonempty cells. Put

$$p_i = q(A_i), \qquad q^i = q(\,\cdot \mid A_i), \qquad p = (p_1, \dots, p_m).$$

If $C_{\mathcal{P}}$ reports only the cell, undetectable-distinctions neutrality gives

$$(q, C_{\mathcal{P}}) \sim (p, \mathrm{Id}_{\{1,\dots,m\}}).$$

The cross-prior bridge therefore identifies the coarse value as

$$G_q(\mu_{q,C_{\mathcal{P}}}) = H_G(p). \tag{BG}$$

Append full revelation within each cell. The final law is χ_q, and the branch formula together with (AS) gives

$$H_G(q) = H_G(p) + Z(q) \sum_i p_i \frac{H_G(q^i)}{Z(q^i)}. \tag{GR}$$

Here singleton cells contribute zero.

Set $w(q) := 1/Z(q) > 0$. Because $\kappa \neq 0$ in the present paragraph, $\kappa H_G(x) = Z(x) - 1$. Substitution into (GR) gives

$$Z(q) - 1 = Z(p) - 1 + Z(q) \sum_i p_i \left(1 - \frac{1}{Z(q^i)}\right).$$

Cancelling and inverting yields the positive grouping-weight equation

$$w(q) = w(p) \sum_i p_i w(q^i). \tag{W}$$

All distributions in what follows are read on their positive-probability supports. For $u \otimes v$, partition by the first coordinate: the coarse distribution is u and every conditional distribution is v. Thus (W) gives

$$w(u \otimes v) = w(u)w(v). \tag{M}$$

Two groupings force the weight to be constant. Take arbitrary finite probability vectors u, v and write $x = w(u)$ and $y = w(v)$. Let T be the law of a triple (ℓ, z_1, z_2): ℓ is uniform on $\{0, 1\}$, and conditional on $\ell = 0$ the pair (z_1, z_2) has law $u \otimes u$, while conditional on $\ell = 1$ it has law $v \otimes v$. In tagged notation,

$$T := \tfrac{1}{2}(u \otimes u)^0 \sqcup \tfrac{1}{2}(v \otimes v)^1.$$

Grouping first by ℓ and using (M) gives

$$w(T) = w(\tfrac{1}{2}, \tfrac{1}{2}) \frac{x^2 + y^2}{2}. \tag{E1}$$

Instead group first by (ℓ, z_1). That marginal is $S = \frac{1}{2}u^0 \sqcup \frac{1}{2}v^1$, so

$$w(S) = w(\tfrac{1}{2}, \tfrac{1}{2}) \frac{x + y}{2}.$$

The conditional law of z_2 is u when $\ell = 0$ and v when $\ell = 1$. A second use of (W) therefore gives

$$w(T) = w(S) \frac{x + y}{2} = w(\tfrac{1}{2}, \tfrac{1}{2}) \left(\frac{x + y}{2}\right)^2. \tag{E2}$$

The common leading factor is positive. Comparing (E1) and (E2) yields $(x - y)^2 = 0$. Thus w is constant on all finite probability vectors. Its singleton value is one, so $w \equiv 1$.

It follows that $Z \equiv 1$, hence $\kappa H_G(q) = 0$ for every q. Axiom (A4) supplies a nondegenerate q with $H_G(q) > 0$, so $\kappa = 0$, contradicting the temporary alternative. Equation (AS) now gives $a_q = C$ universally. Dividing the product and branch formulas by C proves (PA) and (CR′). Since the division is by one common positive number, it also preserves the cross-prior bridge. $\square$

Entropy identification.

Lemma 32 (Entropy reduction and Faddeev identification)**.** *Define the value of full revelation by*

$$\mathcal{H}(q) := V_q(\chi_q),$$

so $\mathcal{H}(q) = H_G(q)/C$, with boundary priors read on their support. Then every finite channel $P : A \to \Delta(X)$ satisfies

$$V_q(\mu_{qP}) = \mathcal{H}(q) - \sum_{x:m_{qP}(x)>0} m_{qP}(x)\mathcal{H}(r_x^{q,P}). \tag{ER}$$

Moreover,

$$\mathcal{H}(q) = \alpha H(q)$$

for a single $\alpha > 0$. Consequently

$$V_q(\mu_{qP}) = \alpha I_{qP}(A; X). \tag{MI}$$

Proof. Entropy reduction. Assume first that q has full support. Append full revelation after P. The final posterior law is χ_q, while the continuation after record x has full-revelation law χ_{r_x} on the support of r_x. The exact chain rule (CR′) therefore gives

$$\mathcal{H}(q) = V_q(\mu_{qP}) + \sum_{x:m_{qP}(x)>0} m_{qP}(x)\mathcal{H}(r_x),$$

which is (ER). A boundary prior is reduced to its support; null action rows and null records affect neither side.

Strong additivity. Let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition of the support of q, and put

$$p_i = q(A_i), \qquad q^i = q(\,\cdot \mid A_i), \qquad p = (p_1, \dots, p_m).$$

The channel $C_{\mathcal{P}}$ which reports the cell is neutral to the corresponding full revelation on the cell index:

$$(q, C_{\mathcal{P}}) \sim (p, \mathrm{Id}_{\{1,\dots,m\}})$$

by Lemma 24. The common-scale cross-prior bridge from Lemma 31 therefore gives

$$V_q(\mu_{q,C_{\mathcal{P}}}) = \mathcal{H}(p).$$

Applying (ER) to $C_{\mathcal{P}}$ yields

$$\mathcal{H}(q) = \mathcal{H}(p) + \sum_i p_i \mathcal{H}(q^i). \tag{FA}$$

Thus $\mathcal{H}$ is strongly additive. Relabelling invariance of V makes $\mathcal{H}$ symmetric; support restriction makes it expansive under adding zero-probability states; and $\mathcal{H}(\delta_a) = 0$. Record processing makes every experiment weakly preferable to its no-information garbling, so $V_q(\mu) \geq 0$ on every fibre and hence $\mathcal{H}(q) \geq 0$. Axiom (A4) gives strict positivity at every nondegenerate full-support prior. Although (FA) was derived by partitioning the positive support, it also holds with zero outer weights: delete the null cells by support restriction, apply the displayed recursion, and reinsert them by expansibility.

Continuity: the binary case. Write $h(t) := \mathcal{H}(t, 1-t)$. We derive, rather than assume, its continuity from Axiom (A2). Fix $c \geq 0$. Choose a full-support binary prior θ with $\mathcal{H}(\theta) > 0$, as supplied by Axiom (A4). Choose $n \geq 1$ and $\lambda \in [0, 1]$ so that

$$c = \lambda n \mathcal{H}(\theta).$$

For $\theta_n := \theta^{\otimes n}$, product additivity (PA) gives $\mathcal{H}(\theta_n) = n\mathcal{H}(\theta)$. Let $E_{n,\lambda}$ reveal the n-bit action fully with probability λ and report a common erasure symbol otherwise. Its posterior law is $\lambda\chi_{\theta_n} + (1-\lambda)\delta_{\theta_n}$, so affinity gives

$$V_{\theta_n}(\mu_{\theta_n,E_{n,\lambda}}) = c.$$

Now fix the block environment

$$K_c := \mathrm{Id}_{\{0,1\}} \sqcup E_{n,\lambda}.$$

Let q_t^0 put the binary prior $(t, 1-t)$ on its first block and let θ_n^1 put θ_n on its second block. The cross-prior bridge says

$$\begin{aligned} h(t) \geq c \quad &\iff \quad q_t^0 \succeq_{K_c} \theta_n^1, \\ h(t) \leq c \quad &\iff \quad \theta_n^1 \succeq_{K_c} q_t^0. \end{aligned}$$

The environment is fixed and $t \mapsto q_t^0$ is continuous, so Axiom (A2) makes both level sets closed. When $c < 0$, the superlevel set is all of $[0,1]$ and the sublevel set is empty because $h \geq 0$. Thus every upper and lower level set of h is closed, and h is continuous.

Continuity on every finite simplex. Proceed by induction on the number of coordinates. For $q = (q_1, (1-q_1)\bar{q})$ with $q_1 < 1$, recursion (FA) for the partition $\{\{1\}, \{2, \ldots, n\}\}$ gives

$$\mathcal{H}(q) = h(q_1) + (1-q_1)\mathcal{H}(\bar{q}). \tag{28}$$

The right side is continuous away from $q_1 = 1$ by binary continuity and the induction hypothesis. At $q_1 = 0$, support restriction and $h(0) = 0$ give the same formula on the face. At $q_1 = 1$, the induction hypothesis makes $\mathcal{H}$ bounded on the compact $(n-1)$-simplex, so the second term in (28) tends to zero and the first tends to $h(1) = 0$. Relabelling handles the other faces. Hence $\mathcal{H}$ is continuous on each closed finite simplex.

We have now verified all hypotheses of the strong-additivity form of Faddeev's theorem: continuity on each finite simplex, invariance under bijections, expansibility, vanishing on point masses, and (FA). The finite entropy characterisation [8, Theorem 6] therefore gives

$$\mathcal{H}(q) = \alpha H(q)$$

for some $\alpha \geq 0$. Axiom (A4) forces $\alpha > 0$. Substitution in (ER) gives

$$V_q(\mu_{qP}) = \alpha \left(H(q) - \sum_x m_{qP}(x) H(r_x^{q,P}) \right) = \alpha I_{qP}(A; X),$$

where the sum is over positive-mass records. This proves (MI). □

Completion of Proposition 18. After Lemma 31, the cross-prior bridge is represented by V. Lemma 32 identifies this value as αI with one $\alpha > 0$. Hence, for arbitrary pairs (q, P) and (r, Q),

$$q^0 \succeq_{P \sqcup Q} r^1 \quad \iff \quad I_{qP} \geq I_{r,Q}, \tag{BMI}$$

including boundary priors by support restriction. For one fixed channel, the duplication clause of Axiom (A5) identifies $q \succeq_P q'$ with the left side of (BMI) for $Q = P$, proving the first assertion. For the block clause, Axiom (A5) deletes all irrelevant blocks and (BMI) compares the two remaining pairs. A lottery supported on block i has a constant block tag, so

$$I_{q_i^i, \bigsqcup_{k \in J} P_k} = I_{q_i P_i},$$

and similarly for block j. This is exactly the stated common-scale block comparison. □

B Completion of the representation theorem

Appendix A proves the one substantial auxiliary input: the pure-trace lemma on the constant-payoff domain. This appendix combines that result with the affine material scale, calibrates deterministic payoff branches, and completes Theorem 1. We also use the ordinary mixture-space theorem: a complete and transitive mixture order satisfying independence and segment calibration has an affine representation.

The logical order is therefore

$$\begin{tikzcd}[column sep=small]
& \boxed{\text{Axioms (A1)–(A8)}} \arrow[dl] \arrow[dr] & \\
\boxed{\begin{array}{c}\text{terminal-law quotient and}\\ \text{affine material scale}\end{array}} \arrow[dr] & & \boxed{\begin{array}{c}\text{pure-trace lemma (Appendix A) and}\\ \text{one common trace scale}\end{array}} \arrow[dl] \\
& \boxed{\text{branch calibration and payoff telescope}} \arrow[d, Rightarrow] & \\
& \boxed{\text{sequentialisation and Theorem 1}}. &
\end{tikzcd}$$

The left branch identifies the value of terminal payoffs and fixes a common material scale. The right branch is proved once on the constant-payoff domain. They meet in the deterministic branch calculation, which aligns material increments with the trace scale and telescopes across all reached branches.

B.1 Preliminaries

We first dispose of three pieces of bookkeeping. They will be used without further comment.

Recall that $(q, K) \succeq (p, L)$ denotes the paper's canonical two-block comparison. For channels E, G at a common prior q, write $E \succeq_q G$ for $(q, E) \succeq (q, G)$. A *finite experiment at* q is a finite payoff–record channel paired with that prior; its marked terminal law is denoted by η_E.

Lemma 33 (Blocks, supports, and dummies)**.** *Suppose Axioms* (A1)*,* (A2)*, and* (A5)–(A7) *hold.*

(a) *After identifying alternatives related by invertible action or record relabellings, the derived comparison of pairs is complete and transitive. If either member of a pair comparison is replaced by an indifferent alternative, the comparison is unchanged.*

(b) *If* $S = \operatorname{supp} q$*, then* (q, K) *is indifferent to the restriction of* (q, K) *to* S*.*

(c) *If* $s \in \Delta(B)$ *and* $K^{\uparrow}(o, r \mid a, b) := K(o, r \mid a)$*, then*

$$(q \otimes s, K^{\uparrow}) \sim (q, K). \tag{29}$$

Proof. Place any finite collection of alternatives in one finite block environment. Axiom (A5) removes unused blocks, and Axiom (A1) supplies completeness and transitivity there. Invertible action and record relabellings are neutral by applying Axioms (A6) and (A7) in both directions. In particular, swapping both tags identifies the two orientations of a two-block comparison. This proves part (a); a three-block environment gives transitivity and a four-block environment gives replacement.

For part (b), process an action in S by its inclusion in A. In the other direction, output $a \in S$ when the realised action is a, and choose an arbitrary element of S off the support. Both processors satisfy (1); rows attached to null processed actions may be completed arbitrarily. Axiom (A7) gives the two comparisons.

For part (c), first project (a, b) to a. Conversely, given a, draw b independently from s and output (a, b). Again the two joint-law identities in (1) are exact, so Axiom (A7) applies in both directions. Part (a) completes both arguments. ∎

Fix for the moment a full-support prior $q \in \Delta(A)$. For a joint channel K, write

$$p_K(o,r) := \sum_a q(a)K(o,r \mid a), \qquad \pi^K_{or}(a) := \frac{q(a)K(o,r \mid a)}{p_K(o,r)}$$

when $p_K(o,r) > 0$, and define its *marked terminal law*

$$\eta_K := \sum_{(o,r):p_K(o,r)>0} p_K(o,r)\,\delta_{(o,\pi^K_{or})}. \tag{30}$$

It records the terminal payoff and the posterior about the realised action. Write $I_q(E)$ for the mutual information between the action and the complete visible output of E under q. When the output is a first-stage record Y, we also write this quantity as $I_q(A;Y)$.

Lemma 34 (Terminal laws and affine fibres)**.** *Under Axioms* (A1)*,* (A2)*, and* (A5)–(A8)*, the following statements hold.*

(a) *If q has full support and $\eta_K = \eta_L$, then $(q,K) \sim (q,L)$.*

(b) *At every full-support q, the quotient of finite experiments by their marked terminal laws has an affine representation W_q.*

Proof. We give the finite Blackwell argument behind part (a). For a payoff o and posterior π, let

$$M_K(o,\pi) := \sum_{\substack{r:\,p_K(o,r)>0\\ \pi^K_{or}=\pi}} p_K(o,r).$$

Equality of marked laws gives $M_K(o,\pi) = M_L(o,\pi) =: M(o,\pi)$. For $p_K(o,r) > 0$, draw a record s of L according to

$$T(s \mid o,r) := \mathbf{1}_{\{p_L(o,s)>0,\,\pi^L_{os}=\pi^K_{or}\}}\frac{p_L(o,s)}{M(o,\pi^K_{or})}. \tag{31}$$

Choose an arbitrary probability row when $p_K(o,r) = 0$. This is a payoff-preserving record processor. Bayes' formula gives

$$q(a)K(o,r \mid a) = p_K(o,r)\pi^K_{or}(a).$$

For a reached target record s, with posterior $\pi = \pi^L_{os}$, it follows that

$$\sum_r K(o,r \mid a)T(s \mid o,r) = \frac{p_L(o,s)}{M(o,\pi)}\frac{\pi(a)}{q(a)}M(o,\pi) = L(o,s \mid a).$$

If s is unreached, full support makes both sides zero. Thus postprocessing K by T gives L. The symmetric construction gives a processor from L to K. Axiom (A6) and Lemma 33(a) give indifference. For a boundary prior, Lemma 33(b) first deletes the null actions.

For part (b), take the quotient of actual finite marked experiments by equality of (30). If E and G are two such experiments, their public mixture first draws a coin independent of the action, runs the selected experiment, and retains the coin in the record. Its marked law is

$$\eta_{tE\oplus(1-t)G} = t\eta_E + (1-t)\eta_G. \tag{32}$$

Axiom (A8) gives, for $0 < t < 1$ and any third experiment L,

$$E \succeq_q L \quad\Longleftrightarrow\quad tE \oplus (1-t)G \succeq_q tL \oplus (1-t)G. \tag{33}$$

For the reverse implication, if $E \not\succeq_q L$, completeness gives $L \succ_q E$; the strict part of Axiom (A8) makes the reverse public mixture strict. If record alphabets differ, put them in a common disjoint union before invoking the axiom; part (a) identifies the result with (32).

It remains only to check the Archimedean step. A closed segment between two experiments, and any fixed comparator, can be realised on one fixed union of their finite record alphabets. Its channel entries vary continuously with $t \in [0, 1]$. Axiom (A2) therefore makes both contour sets along the segment closed. Completeness says that they cover $[0, 1]$; when a target lies between the two endpoints, the sets contain opposite endpoints and connectedness makes them intersect. Thus every such target is indifferent to a point of the segment. The mixture-space theorem now supplies an affine representative W_q. This quotient is closed under every mixture and embedding used below. □

B.2 The two common scales

Lemma 35 (Material utility and a common affine scale)**.** *Under Axioms* (A1)–(A3) *and* (A5)–(A8)*, there are outcomes o^+, o^- and a nonconstant function $u : O \to \mathbb{R}$ such that*

$$u(o^+) = 1, \qquad u(o^-) = 0.$$

The payoff-lottery order is represented by $\sum_o \ell(o)u(o)$. The representatives in Lemma 34 may be normalised so that

$$W_q(\ell) = \sum_o \ell(o)u(o). \tag{34}$$

With this normalisation, adjoining a full-support independent dummy to a full-support prior preserves value, and for full-support q, q',

$$(q, E) \succeq (q', G) \iff W_q(E) \geq W_{q'}(G). \tag{35}$$

Proof. On a singleton action set, terminal-law sufficiency identifies an experiment with its payoff lottery. Public-mixture independence and continuity, proved as in Lemma 34, give an affine utility on $\Delta(O)$. By Lemma 17, Axiom (A3) supplies $o^+ \succ o^-$. Normalise their values to one and zero and let $u(o)$ be the value of the degenerate lottery at o. Affinity on the finite simplex gives expected utility.

An action processor may ignore its input and draw an arbitrary target prior. Axiom (A7), applied in both directions, therefore makes this payoff lottery order independent of the prior and action alphabet. Normalise each W_q at the same two deterministic payoffs. The payoff-lottery embedding is affine and order-reflecting, so positive-affine uniqueness gives (34).

Dummy lifting between full-support fibres is likewise an affine order embedding. Positive-affine uniqueness initially permits

$$W_{q\otimes s}(E^{\uparrow}) = cW_q(E) + d, \qquad c > 0.$$

The two payoff anchors have values zero and one on both sides; hence $c = 1$ and $d = 0$. To compare E at q with G at q', lift each to the common product prior $q \otimes q'$, using the other prior as the dummy. Equation (29), the replacement part of Lemma 33, and the common product representative give (35). □

Let o_* be the outcome fixed in Appendix A to witness Axiom (A4). We now use Proposition 18 on the constant-o_* lift

$$K_P(o, r \mid a) = \mathbf{1}_{\{o=o_*\}} P(r \mid a).$$

This lift commutes with block formation, record processing, action processing, limits on each finite simplex, and compounding. Thus Axiom (A1) gives the pure weak order, while Lemma 17 turns Axiom (A4) into its positive orientation; Axiom (A2) gives continuity; Axiom (A5) gives block coherence; Axioms (A6) and (A7) give the two data processing conditions, including every null-row completion; and Axiom (A8) gives branchwise continuation. These are precisely the hypotheses of Proposition 18. Its conclusion is that every within-channel and block-supported comparison between such lifts is ordered by $I_{qP}(A; R)$.

Lemma 36 (The trace scale)**.** *Under Axioms* (A1)–(A8)*, there is one number $\lambda > 0$, independent of the prior and all finite alphabets, such that every full-support, nontrivial action fibre and every experiment E that pays o_* surely satisfy*

$$W_q(E) = u(o_*) + \lambda I_q(E). \tag{36}$$

Proof. Fix a nontrivial full-support q. On the constant-o_* face, W_q is affine by Lemma 34. For any experiment E, the finite entropy identity

$$I_q(E) = H(q) - \sum_{(o,\pi)\in \operatorname{supp} \eta_E} \eta_E(o,\pi)H(\pi). \tag{37}$$

shows that mutual information depends only on the marked terminal law and is affine under public mixtures. Proposition 18 says that W_q and I_q represent the same nonconstant order on the constant-o_* face. Positive-affine uniqueness therefore gives

$$W_q(K_P) = \lambda_q I_{qP} + b_q, \qquad \lambda_q > 0.$$

The uninformative experiment has mutual information zero and, by (34), value $u(o_*)$. Thus $b_q = u(o_*)$.

The coefficient does not depend on q. Let q and q' be nontrivial full-support priors. Lift full revelation at q to $q \otimes q'$ by adding an independent dummy coordinate. Normalised value and mutual information are both unchanged. Since the common information value is $H(q) > 0$, cancellation gives $\lambda_{q\otimes q'} = \lambda_q$. Lifting full revelation from the other coordinate gives $\lambda_{q\otimes q'} = \lambda_{q'}$. Hence all such coefficients agree. Equivalently, fix the uniform prior on a two-point dummy alphabet and call its coefficient λ. Axiom (A4) fixes the strict orientation, so $\lambda > 0$.

Finally, any experiment that pays o_* surely is a pure record experiment after the constant payoff coordinate is deleted; restoring that coordinate changes neither its marked terminal law nor its mutual information. This proves (36). □

B.3 The branch calculation

Let q be full support on a nontrivial A and let $P : A \to \Delta(Y)$ be a first-stage record channel. Put

$$m_y := \sum_a q(a)P(y \mid a).$$

For a deterministic payoff profile $g : Y \to O$, let $C(P, g)$ denote the compound that first runs P and then pays $g(y)$ with no further record. Write g^o_y for the profile that pays o at y and o_* elsewhere.

Lemma 37 (The reached-branch payoff increment)**.** *For every $y \in Y$ and $o \in O$,*

$$W_q(C(P, g^o_y)) - W_q(C(P, o_*)) = m_y[u(o) - u(o_*)]. \tag{38}$$

Proof. If $m_y = 0$, changing the payoff on branch y does not change the marked terminal law, so both sides are zero. Suppose $m_y > 0$, and put

$$r_y(a) := \frac{q(a)P(y \mid a)}{m_y}, \qquad S_y := \operatorname{supp} r_y.$$

Let $\bar{r}_y$ be the full-support restriction of r_y to S_y. For a marked continuation E on S_y, extend E to A through a support projection, insert it at branch y, and put the payoff o_* for sure at every other branch. Denote the resulting experiment by $J_y(E)$. If $a \notin S_y$, then $r_y(a) = 0$; since $q(a) > 0$ and $m_y > 0$, the definition of r_y gives $P(y \mid a) = 0$. Hence the continuation rows chosen outside S_y are never used on branch y. Support restriction identifies their local versions as well, so the extension is independent of the chosen projection.

Support restriction and Axiom (A8), including its strict part, give

$$E \succeq_{\bar{r}_y} G \quad\Longleftrightarrow\quad J_y(E) \succeq_q J_y(G).$$

For the reverse implication, failure of the local weak comparison gives a strict reverse comparison by completeness, and strict branchwise consistency gives a strict reverse compound comparison. Different continuation record alphabets are first put in a common disjoint union. Moreover, J_y commutes with public mixtures at the level of marked terminal laws. Thus affine uniqueness gives constants $c_y > 0, d_y$ such that

$$W_q(J_y(E)) = c_y W_{\bar{r}_y}(E) + d_y. \tag{39}$$

Suppose first that S_y has at least two elements. At payoff o_*, compare full revelation, E^{id}, with an uninformative record, E^0. Lemma 36 gives the local difference

$$W_{\bar{r}_y}(E^{\mathrm{id}}) - W_{\bar{r}_y}(E^0) = \lambda H(\bar{r}_y).$$

The finite mutual-information chain rule for insertion is

$$I_q(J_y(E)) = I_q(A;Y) + m_y I_{\bar{r}_y}(E). \tag{40}$$

Support extension does not change the last term. Both inserted experiments pay o_* surely, so Lemma 36 and (40) make their aggregate value difference $\lambda m_y H(\bar{r}_y)$. Equation (39) makes the same difference $c_y \lambda H(\bar{r}_y)$. Since $\lambda > 0$ and a nontrivial full-support distribution has positive entropy, $c_y = m_y$.

Locally, deterministic payoffs o and o_* have values $u(o)$ and $u(o_*)$ by (34). Subtracting their two instances of (39), and identifying the resulting insertions with the two deterministic compounds in the statement, proves (38).

If S_y is a singleton, append an independent full-support two-point dummy action to the first-stage action. The reached posterior becomes the product of r_y and the dummy prior and hence has nontrivial support. The branch mass is still m_y, and Lemma 35 says that both compound values are unchanged. Apply the preceding argument in the product problem and remove the dummy. ☐

Lemma 38 (Deterministic branch assembly)**.** *For every deterministic payoff profile $g : Y \to O$,*

$$W_q(C(P,g)) = \lambda I_q(A;Y) + \sum_y m_y u(g(y)). \tag{41}$$

Proof. First observe that the increment from changing one branch is independent of the payoffs on the other branches. Indeed, for any two backgrounds g, h, the two crossed public half-mixtures

$$\tfrac{1}{2}C(P, g[y \mapsto o]) \oplus \tfrac{1}{2}C(P, h[y \mapsto o_*])$$

and

$$\tfrac{1}{2}C(P, g[y \mapsto o_*]) \oplus \tfrac{1}{2}C(P, h[y \mapsto o])$$

have the same marked terminal law. Away from y they contain the same two payoff-posterior atoms; at y they contain o and o_* with the same mass $m_y/2$. Affinity of W_q and cancellation of the two halves show that the two background-specific increments are equal. Taking h to be the constant-o_* profile and applying Lemma 37, then changing the finitely many branches one at a time, gives

$$W_q(C(P,g)) - W_q(C(P,o_*)) = \sum_y m_y \big[u(g(y)) - u(o_*)\big]. \tag{42}$$

The constant-o_* compound has exactly the information in the first-stage record. Lemma 36 therefore gives $W_q(C(P,o_*)) = u(o_*) + \lambda I_q(A;Y)$. Substitute this into (42) and use $\sum_y m_y = 1$. ☐

B.4 Proof of the theorem

Proof of Theorem 1. Assume first Axioms (A1)–(A8). Let q have full support on a nontrivial action set, and let $K : A \to \Delta(O \times R)$ be arbitrary. Put $Y = O \times R$ and take as first stage

$$P_K(y \mid a) := K(y \mid a), \qquad y = (o, r).$$

After $y = (o, r)$, pay o for sure and reveal nothing further. Call the resulting compound K^{seq}. From K, the record processor sends (o, r) to the sequential record $((o, r), *)$. In the reverse direction it sends $((o, r), *)$ to r; inconsistent payoff-record pairs are unreached and may be completed arbitrarily. Both processors preserve the actual payoff coordinate. Hence Axiom (A6) gives $(q, K) \sim (q, K^{\mathrm{seq}})$.

Apply Lemma 38 with $g(o, r) = o$. The two identities needed are

$$I_q(A; Y) = I_{qK}(A; O, R)$$

and

$$\sum_{o,r} p_{qK}(o, r)u(o) = \mathbb{E}_{qK}[u(O)],$$

respectively. Together with sequentialisation, they give

$$W_q(K) = \mathbb{E}_{qK}[u(O)] + \lambda I_{qK}(A; O, R). \tag{43}$$

If A is a singleton, mutual information is zero and the marked terminal law is that of the induced payoff lottery. Equation (34) gives (43) directly. If q is on the boundary, put $S = \operatorname{supp} q$ and write (q^S, K^S) for the restriction. Then

$$(q, K) \sim (q^S, K^S), \qquad V(q, K) = V(q^S, K^S). \tag{44}$$

The first identity is Lemma 33(b); the second follows directly from the finite sums defining expected utility and mutual information. Thus no representative W_q is assigned to a boundary fibre.

Restrict two arbitrary alternatives to their supports, apply (35) and (43) there, and use (44) to transport the comparison back. This gives

$$(q, K) \succeq (p, L) \iff V(q, K) \geq V(p, L), \tag{45}$$

with the same u and λ. Axiom (A5), first by duplication and then by deletion of irrelevant blocks, turns (45) into the within-channel representation and the finite-block moreover clause. The construction made u nonconstant and $\lambda > 0$. This proves the forward implication.

Conversely, suppose that a nonconstant u and a number $\lambda > 0$ represent the family by V. Axiom (A1) follows from the order on $\mathbb{R}$. Expected utility and mutual information are jointly continuous in (K, q) on each finite simplex; for the latter, use the entropy formula and $0 \log 0 = 0$. Hence the set of triples (K, q, p) for which $V(q, K) \geq V(p, K)$ is closed, which is Axiom (A2). A block-supported prior has a constant block label, which changes neither term of V; this proves Axiom (A5) and reduces every pair comparison below to the corresponding two values.

A payoff-preserving record processor leaves the payoff marginal unchanged and weakly lowers $I(A; O, R)$ by data processing. This proves Axiom (A6). Under an action processor, equation (1) again leaves the payoff-record marginal unchanged, and the joint law forms the Markov chain

$$(O, R) - A - B.$$

Thus $I(B; O, R) \leq I(A; O, R)$, proving Axiom (A7). This argument uses the undivided joint-law equation and so also covers every completion on a null processed action.

For a compound channel, the two finite chain rules are

$$\mathbb{E}[u(O)] = \sum_{y:m(y)>0} m(y)\mathbb{E}_{q_y K^y}[u(O)],$$
$$I(A;Y,O,R) = I(A;Y) + \sum_{y:m(y)>0} m(y) I_{q_y K^y}(A;O,R).$$

The first-stage term is common to two continuation profiles. Their aggregate value difference is therefore

$$\sum_{y:m(y)>0} m(y)\big[V(q_y, K^y) - V(q_y, L^y)\big].$$

Branchwise weak improvement makes this sum nonnegative, and a strict improvement on a reached branch makes it positive. This is precisely Axiom (A8).

Because u is nonconstant, choose o^+, o^- with $u(o^+) > u(o^-)$. In the two-action channel of Axiom (A3), its two pure lotteries have these values and zero information, so Axiom (A3) holds. For any $o \in O$, form the four-action channel described in Axiom (A4). Its two lotteries pay o surely and have information values $\log 2$ and zero. Since $\lambda > 0$, the required strict comparison holds—indeed, it holds at every o, although Axiom (A4) requires only one. The same block calculation used above gives the moreover clause with the original witnesses u and λ. □

C Auxiliary results

This appendix collects proofs of the separability, uniqueness, sign, independence, and choice-implication results stated in the main text. None is used to prove Theorem 1; separating them keeps Appendices A and B focused on the representation argument.

C.1 Separability, uniqueness, and sign

For $q \in \Delta(A)$ and $K : A \to \Delta(O \times R)$, write $\pi^{qK}_{or}(a) := q(a)K(o, r \mid a)/p_{qK}(o, r)$ when $p_{qK}(o, r) > 0$ for the posterior induced by the visible pair, and $\mu_{qK} := \sum_{(o,r):p_{qK}(o,r)>0} p_{qK}(o, r)\, \delta_{\pi^{qK}_{or}}$ for its distribution.

Corollary 39 (No payoff–trace complementarity)**.** *Suppose Axioms* (A1)–(A8) *hold. Fix A and $q \in \Delta(A)$. If channels K and L satisfy $\mathbb{E}_{qK}[u(O)] = \mathbb{E}_{q,L}[u(O)]$ and $\mu_{qK} = \mu_{qL}$, then $(q, K) \sim (q, L)$.*

Proof. The result follows from Theorem 1 and $I_{qK}(A; O, R) = H(q) - \int H(\pi) d\mu_{qK}(\pi)$. □

Proof of Proposition 2. Choose $o_* \in O$ with $u(o_*) = \underline{u}$. First note that the range of V is $[\underline{u}, \infty)$. The lower bound is immediate. For $q \in \Delta(A)$ and $t \in [0, 1]$, let a constant-o_* record channel reveal the realised action with probability t and otherwise report $*$:

$$E_t(o_*, a' \mid a) = t\mathbf{1}_{a'=a}, \qquad E_t(o_*, * \mid a) = 1 - t.$$

Then $V(q, E_t) = \underline{u} + \lambda t H(q)$. At the uniform lottery on n actions these values fill $[\underline{u}, \underline{u} + \lambda \log n]$; the intervals are unbounded.

For part (i), let W be a common-scale representation and define $\varphi(v) := W(q, K)$ for any pair with $V(q, K) = v$. This is well defined: equality of the two V-values gives indifference in their two-block environment by Theorem 1, and the common-scale property gives equality of the two W-values. A strict inequality between V-values similarly gives a strict inequality between W-values. Thus φ is strictly increasing and $W = \varphi \circ V$. The converse is immediate from Theorem 1.

For part (ii), the expected-utility and mutual-information chain rules give, for any two continuation profiles,

$$V(q, P \triangleright \{K^y\}) - V(q, P \triangleright \{L^y\}) = \sum_{y:m(y)>0} m(y)\big[V(q_y, K^y) - V(q_y, L^y)\big].$$

Hence every $\alpha V + \beta$, with $\alpha > 0$, is common-scale and branch-additive.

Conversely, let the branch-additive common-scale representation be $W = \varphi \circ V$ and define

$$\psi(x) := \varphi(\underline{u} + x) - \varphi(\underline{u}), \qquad x \geq 0.$$

Fix $n \geq 2$ and let q be uniform on n actions. For $x \in [0, \lambda \log n]$, choose a constant-$o_*$ erasure channel whose value is $\underline{u} + x$. In the first continuation profile, use this channel after the first branch of a public coin with probabilities θ and $1 - \theta$, and use the uninformative constant-o_* channel after the second. Let the comparison profile use the uninformative channel after both branches. The two compound values are $\underline{u} + \theta x$ and $\underline{u}$, so (5) gives

$$\psi(\theta x) = \theta \psi(x) \qquad (0 \leq \theta \leq 1).$$

Taking $x = \lambda \log n$ shows that ψ is linear on $[0, \lambda \log n]$. These intervals are nested and unbounded, so $\psi(x) = \alpha x$ on $[0, \infty)$ for one $\alpha > 0$. Therefore $\varphi(v) = \alpha v + \beta$ on the range of V.

Finally, any second trace-tempered index $\widetilde{V}$ is itself branch-additive. Applying part (ii) and then restricting to deterministic payoff outcomes and to constant-payoff record channels gives $\widetilde{u} = \alpha u + \beta$ and $\widetilde{\lambda} = \alpha \lambda$. □

Proof of Corollary 3. Part (i) is Theorem 1. For part (ii), reverse the primitive relation in every environment. The reversed family satisfies Axioms (A1), (A2), (A5), and (A8); Axiom (A3) holds with its two witnessing outcomes interchanged. It also satisfies the forward versions of Axioms (A4), (A6), and (A7). Theorem 1 therefore represents the reversed family by $\mathbb{E}[\widehat{u}(O)] + \widehat{\lambda} I(A; O, R)$ with $\widehat{\lambda} > 0$. Negating that index represents the original family, so it has the asserted form with $u = -\widehat{u}$ and $\lambda = -\widehat{\lambda} < 0$.

For part (iii), collapse the action alphabet of any pair (q, K) to a singleton. Axiom (A7^0) makes the pair indifferent to the resulting one-action channel, whose output law is p_{qK}. Axiom (A6^0) then erases its record without changing value. Thus every pair is indifferent to the one-action channel that delivers precisely its payoff marginal. Axiom (A5) makes these reductions coherent across blocks. On payoff lotteries, Axiom (A8) applied to an action-independent public coin gives mixture independence, Axiom (A2) gives continuity, and Axiom (A3) gives non-triviality. The mixture-space theorem therefore yields a nonconstant expected-utility index. Conversely, expected utility is unaffected by record or action processing and satisfies the three indifference clauses. The positive and negative formulae satisfy their respective clauses by data processing and the two chain rules. ☐

C.2 Independence of the axioms

Proof of Proposition 4. Fix a nonconstant u and write $U(q, K) := \mathbb{E}_{qK}[u(O)]$, $Z := (O, R)$, and $I := I(A; Z)$. All scores below are extended to block and compound channels by the same formula.

Failure of Axiom (A1). Use the Pareto relation on the two coordinates (U, I):

$$(q, K) \succeq (p, L) \iff U(q, K) \geq U(p, L) \text{ and } I(q, K) \geq I(p, L).$$

It is transitive but incomplete whenever one alternative has more payoff and the other more information. Its graph is closed. Both data-processing axioms are coordinatewise monotonic, and both chain rules make branchwise weak and strict improvements aggregate. Block coherence holds. For Axiom (A3), choose the two outcomes so that $u(o^+) > u(o^-)$; for Axiom (A4), the two test lotteries have equal payoff value and information values $\log 2$ and zero. Thus both relevance axioms hold.

Failure of Axiom (A2). Order alternatives lexicographically by (U, I), with payoff first. This is a weak order and satisfies all structural, processing, and continuation requirements. Axiom (A3) holds through the first coordinate, and Axiom (A4) through the second when the first is tied. It is not continuous even within one fixed channel. Let a three-action channel reveal the action fully and let only action 2 pay the high outcome. Put

$$p = (1/4, 1/2, 1/4), \qquad q = (0.49, 0.50, 0.01), \qquad q_n = (0.49 - \varepsilon_n, 0.50 + \varepsilon_n, 0.01),$$

where $\varepsilon_n \downarrow 0$. The lotteries p and q have the same expected utility but $H(q) < H(p)$, so $p \succ_K q$. Every q_n has strictly higher expected utility than p, so $q_n \succ_K p$. Thus the graph in Axiom (A2) is not closed.

Failure of Axiom (A3). Use $V := I$. All other axioms hold. In particular, the test in Axiom (A4) has information values $\log 2$ and zero. But the two pure lotteries in Axiom (A3) both have zero information, whatever their payoffs.

Failure of Axiom (A4). Use $V := U$. All other axioms hold. The outcomes in Axiom (A3) can be chosen to have different utility, but the two lotteries in Axiom (A4) always have the same payoff and hence are indifferent.

Failure of Axiom (A5). Treat block labels as syntactic tags, with the outermost tag taking precedence. Put $w(a^0) = 1$, $w(a^i) = 0$ for $i \neq 0$, and $w(a) = 0$ on untagged actions. For $\varepsilon > 0$, use

$$V_\varepsilon(q, K) := U(q, K) + I(q, K) + \varepsilon \sum_a q(a) w(a).$$

Every record- or action-processing comparison places the original alternative in block 0, so the data-processing inequality receives the additional ε. If Δ_y is the base-score difference in continuation branch y, its block comparison is weak exactly when $\Delta_y+\varepsilon \geq 0$, while the aggregate block difference is

$$\sum_y m(y)\Delta_y + \varepsilon = \sum_y m(y)(\Delta_y + \varepsilon).$$

Thus Axiom (A8), including strictness, holds; the remaining axioms are immediate. Duplication coherence fails. The two fixed relevance tests are untagged, so their scores reduce to $U + I$ and both strict comparisons hold. Take constant payoff, K revealing a binary action, a degenerate q, and a nearby nondegenerate p with $0 < H(p) < \varepsilon$. Then $p \succ_K q$, but $q^0 \succ_{K \sqcup K} p^1$.

Failure of Axiom (A6). For $\varepsilon > 0$, use

$$V(q,K) := U(q,K) + I(A;Z) - \varepsilon H(Z \mid A). \tag{46}$$

This score is continuous and branch-additive, since both mutual information and conditional entropy obey the chain rule. Under action processing $A \to B$, the score difference between the original and processed alternatives is

$$[I(A;Z) - I(B;Z)] + \varepsilon[H(Z \mid B) - H(Z \mid A)] = (1+\varepsilon)I(A;Z \mid B) \geq 0,$$

so (46) satisfies Axiom (A7). Axiom (A3) holds through U. In the Axiom (A4) test, the first lottery has $I = \log 2$ and $H(Z \mid A) = 0$, while the second has $I = 0$ and $H(Z \mid A) = \log 2$; hence the first is strictly preferred. Record data processing fails: with constant payoff and a fair record independent of A, the original value is $-\varepsilon \log 2$; deleting that noise gives value zero.

Failure of Axiom (A7). Let $G(q) := 1 - \sum_a q(a)^2$ be Gini entropy and define its posterior reduction

$$J_G(q,K) := G(q) - \int G(\pi)\, d\mu_{qK}(\pi), \qquad V := U + J_G.$$

Concavity of G gives record data processing, and the potential-difference form gives the exact branch chain rule. Axiom (A3) holds through U. In the Axiom (A4) test, J_G is $1/2$ under the first lottery and zero under the second, so trace relevance holds. Action processing can nevertheless increase J_G. Take $q = (1/8, 1/8, 3/4)$, a binary record with

$$\Pr(0 \mid a_1) = \Pr(0 \mid a_2) = 0, \qquad \Pr(0 \mid a_3) = 1/4,$$

and merge a_1, a_2 into one processed action. Direct calculation gives $J_G = 9/416$ before the merge and $J_G = 3/104$ afterwards. Hence Axiom (A7) fails.

Failure of Axiom (A8). Use $V := U + \min\{I, 1\}$, with information measured in bits. It is continuous, respects both data-processing axioms, and satisfies block coherence. Axiom (A3) holds through U, while the test in Axiom (A4) has information values one and zero. Let q be uniform on four actions. A first-stage record reveals which of two pairs contains the action. In every branch, revealing the action within the pair strictly improves the continuation from zero to one bit. The two compounds have respectively one and two bits before capping and are therefore indifferent. The strict clause of Axiom (A8) fails. □

C.3 Proofs of the choice implications

Results stated in the text only in summary form (Propositions 40, 44, and 41, Corollary 42, and Remark 47) are restated here in full. Proofs appear in an order that front-loads shared machinery.

Proof of Proposition 5. (i) Under either lottery, each visible pair (o_0, r_j), $j = 1, 2$, has probability $\frac{1}{2}$: under q' because each a_j produces (o_0, r_j) for sure, under q'' because each of a_3, a_4 produces either pair with equal probability. Hence $p_{q'K} = p_{q''K}$. On $\operatorname{supp}(q')$ the rows are degenerate and distinct, so the visible pair reveals the action and $I_{q'K} = H(q') = \log 2$; on $\mathrm{supp}(q'')$ the rows are identical, so the visible pair is independent of the action and $I_{q''K} = 0$. The payoff terms agree because the payoff marginals do, giving the displayed gap. (ii) All rows of K° are equal, so $I_{qK^\circ} = 0$ for every q; since the payoff is constantly o_0, the representation gives indifference. □

Proof of Proposition 7. Expected utility is linear in q, while mutual information is concave in the input distribution of a fixed channel [17]. Continuity on the compact simplex gives existence. Using

$$I_{qK}(A;Z) = \sum_a q(a) D_{\mathrm{KL}}(K_Z(\cdot \mid a) \,\|\, m_{qK}),$$

its directional derivative in the mass on action a equals the displayed divergence up to a term common to every action. The result is the Kuhn–Tucker condition for a concave programme. At constant u it reduces to the classical capacity condition. □

Proof of Proposition 8. The entropy identity gives

$$V(q,K) = \lambda H(m_{qK}) + \sum_a q(a)\big[\bar{u}(a) - \lambda H(K_a)\big].$$

If q_0 and q_1 are optimal with $m_{q_0K} \neq m_{q_1K}$, strict concavity of Shannon entropy and linearity of $q \mapsto m_{qK}$ and of the remaining sum give $V(tq_0 + (1-t)q_1, K) > C^*(K)$ for $t \in (0,1)$, a contradiction. Hence every optimum induces the same m^*. By Proposition 7, an optimum q^* satisfies the displayed equalities at some constant c; averaging them under q^* gives $c = V(q^*,K) = C^*(K)$, so $\mathrm{supp}(q^*) \subseteq \mathcal{A}^*$. Conversely, if $m_{qK} = m^*$ and $\mathrm{supp}(q) \subseteq \mathcal{A}^*$, then

$$V(q,K) = \sum_a q(a)\big[\bar{u}(a) + \lambda D_{\mathrm{KL}}(K_a \| m^*)\big] = C^*(K).$$

Affine independence makes the representation $m^* = \sum_a q(a) K_a$ with $\mathrm{supp}(q) \subseteq \mathcal{A}^*$ unique. Identical rows share $\bar{u}(a)$ and $H(K_a)$, so both m_{qK} and the displayed expression for V are unchanged by redistribution within a class; collapsing the classes and lifting aggregate masses proves the last claim. □

Proof of Proposition 6. Throughout, m^* denotes the common optimal marginal of Proposition 8. (i) Suppose $m^*(z) = 0$ for some $z \in Z_K$ and choose a with $K_Z(z \mid a) > 0$. Then $D_{\mathrm{KL}}(K_Z(\cdot \mid a)\|m^*) = +\infty$, which is incompatible with the finite on-support equality if some optimum uses a and with the off-support inequality of Proposition 7 if none does. Under the private-consequence condition, an optimum with $q^*(a) = 0$ would force $m^*(z_a) = 0$.

(ii) For $q = \delta_a$ the induced marginal is $K_Z(\cdot \mid a)$ and the on-support constant is $\bar{u}(a)$; the displayed system is the off-support condition of Proposition 7, necessary and sufficient by concavity. Under strict inequalities, $\mathcal{A}^* = \{a\}$ in Proposition 8, so every optimum is δ_a. □

Proposition 40 (Optimal mixing)**.** *Let $A = \{a,b\}$ with $K_a := K_Z(\cdot \mid a) \neq K_b := K_Z(\cdot \mid b)$ and $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$. For $x \in [0,1]$ let q_x put mass x on b, so that $m_{q_xK} = (1-x)K_a + xK_b =: m_x$, and for $x \in (0,1)$ define*

$$g(x) := D_{\mathrm{KL}}(K_b \,\|\, m_x) - D_{\mathrm{KL}}(K_a \,\|\, m_x).$$

- (i) *g is continuous and strictly decreasing, with $g(0^+) = D_{\mathrm{KL}}(K_b \,\|\, K_a)$ and $g(1^-) = -D_{\mathrm{KL}}(K_a \,\|\, K_b)$ as extended-real limits. If $\Delta < \lambda\, D_{\mathrm{KL}}(K_b \,\|\, K_a)$, the optimum is unique and interior, $q^* = q_{x^*}$ with x^* the unique solution of $\Delta = \lambda g(x)$.*
- (ii) *The optimal mass $x^*(\Delta,\lambda)$ on the inferior action is nonincreasing in Δ and nondecreasing in λ, strictly so on the mixing region; at $\Delta = 0$ it is the capacity-achieving input of the two-row channel, independently of λ.*

Proof. Write $f(x) := V(q_x, K) = \bar{u}(a) - x\Delta + \lambda I(x)$ with $I(x) = H(m_x) - (1-x)H(K_a) - xH(K_b)$. Since $K_a \neq K_b$, $x \mapsto m_x$ is affine and injective, so strict concavity of H [17] makes f strictly concave on $[0,1]$, and a unique maximiser exists. On $(0,1)$ every $z \in \operatorname{supp} K_a \cup \operatorname{supp} K_b$ has $m_x(z) > 0$ and, using $\sum_z (K_b(z) - K_a(z)) = 0$,

$$I'(x) = H(K_a) - H(K_b) - \sum_z (K_b(z) - K_a(z)) \log m_x(z) = g(x),$$

so $f'(x) = -\Delta + \lambda g(x)$, with g continuous and strictly decreasing as the derivative of a strictly concave function. As $x \downarrow 0$, $m_x(z) \to K_a(z)$; if $\operatorname{supp} K_b \subseteq \operatorname{supp} K_a$ every summand converges and $g(0^+) = D_{\mathrm{KL}}(K_b \| K_a)$, while if some $\bar{z}$ has $K_b(\bar{z}) > 0 = K_a(\bar{z})$ the summand $-K_b(\bar{z}) \log(x K_b(\bar{z}))$ diverges to $+\infty$, matching the extended-sense divergence. Exchanging the roles of a and b gives the limit at 1. By concavity and the mean value theorem, δ_a is optimal if and only if $\lim_{x\downarrow 0} f'(x) \leq 0$, i.e. $\Delta \geq \lambda\, D_{\mathrm{KL}}(K_b \| K_a)$; this is the threshold stated in the text, and Proposition 6(ii) with two actions. If $\Delta < \lambda\, D_{\mathrm{KL}}(K_b \| K_a)$, the maximiser is not $x = 0$ by the threshold and not $x = 1$ because $f'(1^-) = -\Delta - \lambda D_{\mathrm{KL}}(K_a \| K_b) < 0$; it is therefore the unique root of $\Delta = \lambda g(x)$, which exists because g decreases strictly between its limits. This proves (i). The comparative statics of (ii) follow by inverting the strictly decreasing g at Δ/λ: raising Δ raises the target, raising λ lowers it; at $\Delta = 0$ the condition $g(x) = 0$ is the capacity condition of Proposition 7 at constant payoff. $\square$

Proposition 41 (Trace demand)**.** *Fix K and write $e(q) := \mathbb{E}_{qK}[u(O)]$, $i(q) := I_{qK}(A; O, R)$,*

$$V^*(\lambda) := \max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}, \qquad Q(\lambda) := \arg\max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}.$$

(i) *V^* is convex on $(0,\infty)$, i is affine on the convex compact set $Q(\lambda)$, and*

$$(V^*)'_-(\lambda) = \min_{q \in Q(\lambda)} i(q), \qquad (V^*)'_+(\lambda) = \max_{q \in Q(\lambda)} i(q);$$

at every point of differentiability, $(V^*)'(\lambda) = i(q)$ for every $q \in Q(\lambda)$.

(ii) *Let $0 < \lambda_1 < \lambda_2$ and $q_j \in Q(\lambda_j)$, writing $e_j := e(q_j)$, $i_j := i(q_j)$. Then $i_1 \leq i_2$, $e_1 \geq e_2$, and*

$$\lambda_1(i_2 - i_1) \leq e_1 - e_2 \leq \lambda_2(i_2 - i_1).$$

Hence either $(e_1, i_1) = (e_2, i_2)$, or $i_1 < i_2$ and $e_1 > e_2$ and

$$\lambda_1 \leq \frac{e_1 - e_2}{i_2 - i_1} \leq \lambda_2.$$

Proof of Proposition 41. (i) V^* is a pointwise maximum of functions affine in λ, hence convex and, since $0 \leq i \leq \log|A|$, finite. On the convex compact set $Q(\lambda)$ the objective is constant and e is linear, so $i = (V^* - e)/\lambda$ is affine there. For $q \in Q(\lambda)$ and $\mu > 0$, $V^*(\mu) \geq V^*(\lambda) + (\mu - \lambda) i(q)$, so each such $i(q)$ is a subgradient. For the right derivative, choose $q_h \in Q(\lambda + h)$; then

$$\max_{q \in Q(\lambda)} i(q) \leq \frac{V^*(\lambda + h) - V^*(\lambda)}{h} \leq i(q_h),$$

the first inequality by the subgradient bound, the second by optimality of $Q(\lambda)$ at λ. By compactness and continuity, accumulation points of $(q_h)_{h\downarrow 0}$ lie in $Q(\lambda)$, so the right-hand side accumulates in $i(Q(\lambda))$ and the right derivative equals the maximum. The left derivative is symmetric, and a convex function on an interval is differentiable off a countable set.

(ii) Optimality at the two weights gives $e_1 + \lambda_1 i_1 \geq e_2 + \lambda_1 i_2$ and $e_2 + \lambda_2 i_2 \geq e_1 + \lambda_2 i_1$. Adding yields $(\lambda_2 - \lambda_1)(i_2 - i_1) \geq 0$, hence $i_2 \geq i_1$; the two inequalities rearrange to the displayed bracket, which gives $e_1 \geq e_2$ and, when the pairs differ, the final bound. $\square$

Corollary 42 (Extreme trace weights)**.** *Let $e^0 := \max_q e(q)$, $i^0 := \max\{i(q) : e(q) = e^0\}$, $C := \max_q i(q)$, and $e^C := \max\{e(q) : i(q) = C\}$. If $\lambda_n \downarrow 0$ and $q_n \in Q(\lambda_n)$, every accumulation point of (q_n) maximises i among the maximisers of e; if $\lambda_n \uparrow \infty$, every accumulation point maximises e among the maximisers of i. Moreover*

$$V^*(\lambda) = e^0 + \lambda i^0 + o(\lambda) \quad \text{as } \lambda \downarrow 0, \qquad V^*(\lambda) = \lambda C + e^C + o(1) \quad \text{as } \lambda \uparrow \infty.$$

Proof of Corollary 42. For $\lambda_n \downarrow 0$, comparing q_n with an e-maximiser attaining i^0 gives

$$0 \le e^0 - e(q_n) \le \lambda_n \big(i(q_n) - i^0\big).$$

Boundedness of i forces $e(q_n) \to e^0$, and the same inequality gives $i(q_n) \ge i^0$; continuity delivers the accumulation claim. Dividing by λ_n,

$$\frac{V^*(\lambda_n) - e^0}{\lambda_n} = i(q_n) - \frac{e^0 - e(q_n)}{\lambda_n},$$

where the last fraction lies in $[0, i(q_n) - i^0]$ and vanishes, proving the first expansion. For $\lambda_n \uparrow \infty$, comparing q_n with an i-maximiser attaining e^C gives $0 \le \lambda_n(C - i(q_n)) \le e(q_n) - e^C$, and the symmetric argument completes the proof. □

Proof of Proposition 10. The processor leaves the payoff coordinate unchanged, so the expected-utility terms coincide and the difference is $\lambda[I_{qK}(A; O, R) - I_{qL}(A; O, R')]$. Write $Z := (O, R)$ and $Z' := (O, R')$. By construction $A \to Z \to Z'$ is a Markov chain, so expanding $I(A; Z, Z')$ in both orders gives

$$I(A; Z) - I(A; Z') = I(A; Z \mid Z') = I(A; R \mid O, R'),$$

the last equality because O is part of Z'. Conditional mutual information is nonnegative and vanishes exactly under the conditional independence in (b) [17], proving the identity and (a)⇔(b). For (b)⇒(c): the Markov property gives $\Pr(A = \cdot \mid Z, Z') = \pi^{qK}_Z$ almost surely, while (b) gives $\Pr(A = \cdot \mid Z, Z') = \pi^{qL}_{Z'}$; the two random posteriors coincide almost surely, so their laws are equal. For (c)⇒(a): by the identity in the proof of Corollary 39, the information term depends on (q, K) only through μ_{qK}, and the payoff terms already agree. □

Proof of Corollary 11. Part (i) is the chain rule with an action-independent first stage: $I(A; Y) = 0$, and each reached branch reproduces the joint law of the corresponding constituent. Part (ii) follows from Proposition 10: deleting the coin coordinate is a deterministic payoff-preserving record processor carrying the public mixture to M_t. □

Proof of Proposition 16. (i) If $L = KT$, Proposition 10 gives $I_{qK} \ge I_{qL}$ at every q. Because the payoff kernels coincide, $V(q, K) - V(q, L) = \lambda[I_{qK} - I_{qL}]$ with $\lambda > 0$, so uniform value dominance and $K \succeq_c L$ are the same statement.

(ii) Let $A = \{0, 1\}$, fix $o^* \in O$, and let both channels pay o^* surely. Let K carry the binary erasure record with erasure probability $\varepsilon = \frac{1}{2}$ and L the binary symmetric record with crossover $p = \frac{1}{5}$:

$$K(o^*, a \mid a) = 1 - \varepsilon, \quad K(o^*, \mathrm{e} \mid a) = \varepsilon, \qquad L(o^*, a \mid a) = 1 - p, \quad L(o^*, 1 - a \mid a) = p.$$

The payoff kernels agree row by row, so it suffices to compare informations, and the comparison is invariant to the base; compute in bits with h the binary entropy. For $q = (1 - t, t)$, grouping records gives

$$I_{qK} = (1 - \varepsilon)\, h(t), \qquad I_{qL} = h(p * t) - h(p), \qquad p * t := p(1 - t) + (1 - p)t.$$

By Mrs Gerber's Lemma [46], $v \mapsto h(p * h^{-1}(v))$ is convex on $[0,1]$, with h^{-1} valued in $[0, \frac{1}{2}]$; since $h(p*t)$ is invariant under $t \mapsto 1-t$, take $t \le \frac{1}{2}$ and $v = h(t)$. The chord between $v = 0$ and $v = 1$ gives

$$h(p*t) \le (1-v)\,h(p) + v, \qquad \text{hence} \qquad I_{qL} \le \big(1-h(p)\big)h(t) \le (1-\varepsilon)\,h(t) = I_{qK},$$

the last step because $\varepsilon = \frac{1}{2} \le h(\frac{1}{5})$. Thus $K \succeq_c L$. Now suppose $L = KT$ for a processor T, and write $t_r := T(0 \mid o^*, r)$ for $r \in \{0, 1, \mathrm{e}\}$. Matching the probability of the processed record 0 in the two rows,

$$(1-\varepsilon)t_0 + \varepsilon t_{\mathrm{e}} = 1-p, \qquad (1-\varepsilon)t_1 + \varepsilon t_{\mathrm{e}} = p,$$

and subtracting gives $t_0 - t_1 = (1-2p)/(1-\varepsilon) = \frac{6}{5} > 1$, impossible. Compositions of record processors are record processors, so no chain of garblings produces L from K either. $\square$

The next definition and proposition formalise the side-bet elicitation of §4.4. Let o^+ and o^- attain the maximum and minimum of u, write $\Delta u := u(o^+) - u(o^-) > 0$, and for $\beta \in [0,1]$ let $w(\beta) := \beta u(o^+) + (1-\beta)u(o^-)$.

Definition 43 (Side bet)**.** For $K : A \to \Delta(O \times R)$, $\varepsilon \in [0,1)$, and $\beta \in [0,1]$, let $P_\varepsilon : A \to \Delta(\{0,1\})$ be the public coin $P_\varepsilon(1 \mid a) = \varepsilon$, and let $B_\beta : A \to \Delta(O \times \{*\})$ have the constant rows $B_\beta(o^+, * \mid a) = \beta$ and $B_\beta(o^-, * \mid a) = 1-\beta$. Write

$$D_{\varepsilon,\beta}K := P_\varepsilon \triangleright \{K,\, B_\beta\},$$

as in (2), listing the branch-0 and branch-1 continuations in order.

Proposition 44 (Willingness to pay for trace)**.** *Fix $q \in \Delta(A)$, $p \in \Delta(B)$, and channels $K : A \to \Delta(O \times R)$ and $L : B \to \Delta(O \times S)$.*

(i) *For every $\varepsilon \in [0,1)$ and $\beta \in [0,1]$,*

$$V(q,\, D_{\varepsilon,\beta}K) = (1-\varepsilon)\,V(q,K) + \varepsilon\, w(\beta).$$

(ii) *If $\varepsilon \in (0,1)$ and $\beta, \beta' \in [0,1]$ satisfy $(q, D_{\varepsilon,\beta}K) \sim (p, D_{\varepsilon,\beta'}L)$, then*

$$\frac{\varepsilon}{1-\varepsilon}\,[w(\beta) - w(\beta')] = V(p,L) - V(q,K).$$

When the material parts agree, $\mathbb{E}_{qK}[u(O)] = \mathbb{E}_{p,L}[u(O)]$, the left-hand side equals $\lambda[I_{pL}(B;O,S) - I_{qK}(A;O,R)]$: information is priced linearly at rate λ.

(iii) *Such a pair (β, β') exists if and only if $|V(p,L) - V(q,K)| \le \frac{\varepsilon}{1-\varepsilon}\Delta u$; a boundary response therefore brackets the value difference instead of measuring it.*

Proof. (i) The material part of the compound is $(1-\varepsilon)\mathbb{E}_{qK}[u(O)] + \varepsilon w(\beta)$. For the trace part, the coin is action-independent, so both branch posteriors equal q and the chain rule for the compound gives an information term of $(1-\varepsilon)I_{qK}(A;O,R)$, the side-bet branch contributing zero. (ii) By the moreover clause of Theorem 1, the indifference holds if and only if $(1-\varepsilon)V(q,K) + \varepsilon w(\beta) = (1-\varepsilon)V(p,L) + \varepsilon w(\beta')$; rearrange, and substitute the representation when the material parts cancel. (iii) $w(\beta) - w(\beta')$ is continuous on $[0,1]^2$ with range $[-\Delta u, \Delta u]$; apply the intermediate value theorem to the equation in (ii). $\square$

Lemma 45 (The information–payoff plane)**.** *Write $\phi(q,K) := \big(I_{qK}(A;O,R),\, E_{qK}\big)$, and let $\underline{u} := \min_o u(o)$ and $\overline{u} := \max_o u(o)$. The image of ϕ over all finite pairs (q,K) is exactly $[0,\infty) \times [\underline{u}, \overline{u}]$, and the indifference classes of the block-comparison relation are, in these coordinates, the nonempty intersections of the lines $E = c - \lambda I$ with that strip.*

Proof. Inclusion is immediate. Conversely, fix (t, w) in the strip, choose n with $\log n \geq t$, and on $A = \{1, \ldots, n\}$ use continuity of entropy along a segment from a point mass to the uniform lottery to find q_t with $H(q_t) = t$. Let $K(o^+, a \mid a) = \beta$ and $K(o^-, a \mid a) = 1 - \beta$ with β chosen so that $w(\beta) = w$. The payoff lottery is action-independent with mean w, and the record reveals the action, so $\phi(q_t, K) = (t, w)$. The description of the indifference classes then reads the representation through ϕ, using the moreover clause of Theorem 1. $\square$

Proof of Proposition 12. The identity rearranges $E_{qK} + \lambda I_{qK} = E_{pL} + \lambda I_{pL}$. For uniqueness, let $\lambda' \neq \lambda$, put $t := \frac{1}{2} \min\{1, (\overline{u} - \underline{u})/\lambda\} > 0$, and use Lemma 45 to realise pairs with ϕ-coordinates $(t, \overline{u} - \lambda t)$ and $(0, \overline{u})$; they are indifferent, while $E + \lambda' I$ differs across them by $(\lambda - \lambda')t \neq 0$. $\square$

We restate Proposition 14 with its standing hypotheses. Both families are defined over the same outcome set O, on the common domain of all finite pairs, with mutual information in one fixed base. A channel B is *silent* if all its rows are equal; then the joint law under any p is a product, so $I_{pB} = 0$ and $V^i(p, B)$ is the expectation of u_i under the O-marginal of the common row. Since one-action channels realise every payoff lottery in $\Delta(O)$, silent pairs realise exactly the payoff lotteries.

Proof of Proposition 14. Throughout, block comparisons are read through the moreover clause of Theorem 1.

Suppose the representations are (u, λ_1) and (u, λ_2) with $\lambda_2 \geq \lambda_1$. For any (q, K) and silent (p, B),

$$V^2(q, K) - V^2(p, B) = \left[V^1(q, K) - V^1(p, B)\right] + (\lambda_2 - \lambda_1)I_{qK} \geq V^1(q, K) - V^1(p, B),$$

since $I_{qK} \geq 0$ and the silent pair's value is material under both. A weak (or strict) agent-1 preference therefore transfers to agent 2, and reading the display from right to left gives the mirrored clause $(p, B) \succeq^2 (q, K) \Rightarrow (p, B) \succeq^1 (q, K)$: the weak one-sided definition, its strict variant, and the two-sided variant coincide on this class.

Conversely, suppose $\{\succeq^2_K\}$ is more trace-seeking than $\{\succeq^1_K\}$, with representations (u_1, λ_1) and (u_2, λ_2).

Step 1: common utility. Let $D := \{\mu - \nu : \mu, \nu \in \Delta(O)\}$ and $f_i(d) := \sum_o u_i(o)d(o)$; each $f_i \neq 0$ on D because u_i is nonconstant. Silent pairs realise every payoff lottery, and the definition's left argument ranges over all pairs, so silent-against-silent instances are in its scope: $f_1(d) \geq 0 \Rightarrow f_2(d) \geq 0$ on D. If $f_1(d) = 0$, applying this to d and $-d$ gives $f_2(d) = 0$, so $\ker f_1 \subseteq \ker f_2$ and $f_2 = \alpha f_1$ for some $\alpha \in \mathbb{R}$. If $\alpha = 0$ then u_2 is constant, contradicting Axiom (A3) for family 2. (The case is not idle: an index with constant u_2, which violates only that axiom, is indifferent across all silent pairs and is more trace-seeking than every family whatsoever.) If $\alpha < 0$, pick o^+, o^- with $u_1(o^+) > u_1(o^-)$: the silent instance $\delta_{o^+} \succ^1 \delta_{o^-}$ forces $\delta_{o^+} \succeq^2 \delta_{o^-}$, while $\alpha < 0$ gives the strict reverse. Hence $\alpha > 0$ and $u_2 = \alpha u_1 + \beta$. Since $((u_2 - \beta)/\alpha, \lambda_2/\alpha)$ again represents family 2, choose $u_2 = u_1 =: u$, replacing λ_2 by $\lambda_2/\alpha > 0$. The choice is innocuous: any two common-utility normalisations rescale both weights by one positive factor, because u nonconstant forces the factors to agree, so the order of the weights does not depend on it.

Step 2: weight ordering. Write $\overline{u} := \max_o u(o)$ and $\underline{u} := \min_o u(o)$; $\overline{u} > \underline{u}$ by Axiom (A3). Suppose $\lambda_2 < \lambda_1$ and fix $\lambda' \in (\lambda_2, \lambda_1)$. Choose $t \in (0, (\overline{u} - \underline{u})/\lambda')$ and $w := \underline{u} + \lambda' t \in (\underline{u}, \overline{u})$. By Lemma 45 there is a pair (q, K) with $I_{qK} = t$ and $E_{qK} = \underline{u}$; its payoff lottery is action-independent, so after Step 1 both families assign it the same coordinates. Choose a silent (p, B) whose common-row O-marginal pays o^+ with probability $(w - \underline{u})/(\overline{u} - \underline{u})$ and o^- otherwise, so $V^1(p, B) = V^2(p, B) = w$. Then $V^1(q, K) = \underline{u} + \lambda_1 t > w$, so $(q, K) \succ^1 (p, B)$, and the definition gives $(q, K) \succeq^2 (p, B)$; but $V^2(q, K) = \underline{u} + \lambda_2 t < w$, a contradiction. Hence $\lambda_2 \geq \lambda_1$. $\square$

Remark 46. The proof uses $\lambda_i > 0$ only in the choice $E_{qK} = \underline{u}$; taking $E_{qK} = \overline{u}$ when $\lambda' < 0$ runs both directions verbatim. The equivalence therefore holds across all three orientations of Corollary 3: more trace-seeking and less trace-averse are one comparative.

Remark 47 (Eliciting λ)**.** Normalise $u(o^+) = 1$ and $u(o^-) = 0$. All comparisons below are block comparisons, hence within the primitive domain, and no information quantity is estimated: the information gap is an accounting identity of the design. *Step 1.* For each $o \in O$, elicit β_o with $(\delta_*, K_o) \sim (\delta_*, B_{\beta_o})$, where K_o is the degenerate one-action channel used in Lemma 17; one-action channels carry zero information, so $u(o) = \beta_o$. *Step 2.* Fix $o \in O$, $n \geq 2$, the uniform q on $\{1, \dots, n\}$, a side-bet probability $\varepsilon \in (0, 1)$, and the constant-payoff channels K_o^{id} and K_o^0 used in Lemma 17. Elicit β with

$$(q,\ D_{\varepsilon,\beta} K_o^0) \sim (q,\ D_{\varepsilon,0} K_o^{\mathrm{id}}); \qquad \text{then} \qquad \lambda = \frac{\varepsilon}{1-\varepsilon} \cdot \frac{\beta}{\log n},$$

by Proposition 44(i) applied to $V(q, K_o^{\mathrm{id}}) = u(o) + \lambda \log n$ and $V(q, K_o^0) = u(o)$. Such a β exists if and only if $\lambda \log n \leq \varepsilon/(1-\varepsilon)$; a subject still strictly preferring the right block at $\beta = 1$ reveals $\lambda > \varepsilon/\big((1-\varepsilon)\log n\big)$, and ε is raised. *Controls.* Orientation: Theorem 1 implies $(q, K_o^{\mathrm{id}}) \succ (q, K_o^0)$ at every payoff, although Axiom (A4) assumes the comparison at only one. Garbling: under the total record garbling of K_o^{id}, both blocks carry zero information and indifference must obtain; a surviving strict preference indicates that an uncontrolled feature of the display still traces the action. Transport: repeating Step 2 at another n, prior, or payoff level must return the same λ, by Proposition 12(ii); an estimate that declines in $\log n$ is the signature of the capped-information criterion.

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